

Development depends on the constraints between people and markets

Three rigorous economics papers · two current applied-development studies and one field-experiment classic. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

Across Uganda, Burkina Faso, and Kenya, the binding constraint is rarely just a missing asset. Childcare can free market vendors' time, but location determines take-up and whether gains appear in earnings or family well-being. Electricity reaches relatively few homes, yet changes water systems, schools, communications, and finance for whole communities. Informal kin obligations can operate like a tax on visible income, causing women to conceal resources and forgo returns. The literature is converging on **access plus social structure**: public investments work through delivery choices, spillovers, and household relationships that determine who can use a service and how much of its return they can keep.

01 CURRENT / GENERAL INTEREST · STARTS ON PAGE 2 · World Bank Policy Research Working Paper 11442 · 2026 · randomized evaluation · CC BY 3.0 IGO

Childcare for Informal-Sector Workers: Delivery Models, Access, and Welfare

Can free childcare improve both livelihoods and child development for women already working in Kampala's informal markets, and does delivery location matter? The study randomizes 883 vendors to workplace-based care, community-based care, or control, reporting pre-specified intent-to-treat and take-up-adjusted estimates. Childcare shifts care away from family members, raises hourly earnings, and improves women's well-being and child development. Community centers attract more users and larger welfare gains; workplace centers appear to generate larger economic returns. Earnings estimates are imprecise, follow-up is short, and differential tracking plus one-city implementation temper extrapolation, but the design makes the service-delivery trade-off unusually concrete.

02 CURRENT / INFRASTRUCTURE AND DEVELOPMENT · STARTS ON PAGE 17 · Journal of Development Economics 178 · 2026 · peer reviewed · CC BY 4.0

Community Effects of Electrification: Evidence from Burkina Faso's Grid Extension

Does extending the grid help a community when few households connect? The paper links Burkina Faso's 2008–2017 rollout across 558 localities to satellite, administrative, and household data. Engineering constraints generate quasi-exogenous timing for staggered difference-in-differences, event studies, and supporting cross-sectional IV estimates. Electrification increases nighttime light, piped-water systems, school electrification, appliances, and financial inclusion, with benefits reaching unconnected households. Prioritization of higher-potential places and limited survey waves leave residual selection and precision concerns, but the engineering tests, pre-trends, and alternative specifications support a community-infrastructure channel rather than a household-connection story.

03 CLASSIC / SOCIAL NETWORKS AND DEVELOPMENT · STARTS ON PAGE 31 · Review of Economic Studies 83(1) · 2016 · classic · World Bank open edition

Does Africa Need a Rotten Kin Theorem? Experimental Evidence from Village Economies

Can obligations to share income with kin discourage profitable investment? In 26 rural Kenyan villages, 2,145 participants receive randomized endowments, observability rules, and prices for keeping returns private. Women, especially those with relatives present, sacrifice expected earnings to conceal income; structural estimates imply an informal tax above 4 percent that nearly doubles under direct kin observation. Random assignment identifies the effect of observability, while mixed-logit estimates discipline the mechanism. The village-level development correlations are not causal, the setting is one region, and structural results depend on utility assumptions, but the experiment cleanly reveals a hidden social cost that ordinary surveys struggle to measure.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

Childcare for Informal-Sector Workers

Delivery Models, Access, and Welfare

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Abstract

Childcare burdens limit productivity among women in the informal sector, but little is known about the optimal design of childcare support. The study randomizes 883 women market vendors in Kampala, Uganda to receive free workplace- or community-based childcare. Free childcare redistributes caregiving away from household members, raises women's hourly earnings, and improves

child development and women's well-being. Community centers achieve higher utilization and larger gains in development and well-being, while workplace centers generate larger economic impacts. Both models have a high marginal value of public funds. Community centers currently dominate, but workplace-based centers could prove superior if operated at scale.

This paper is a product of the Development Impact Group and the Development Research Group, Development Economics. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at <http://www.worldbank.org/prwp>. The authors may be contacted at sherskowitz@worldbank.org and mlang@worldbank.org. A verified reproducibility package for this paper is available at <http://reproducibility.worldbank.org>, click **here** for direct access.



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1 Introduction

The informal sector plays a critical role as a source of jobs, employing 58% of workers globally and 84% of workers in Africa. Women are systematically overrepresented in the informal sector.¹ Simultaneously, women are disproportionately charged with the responsibility of providing unpaid work for their families, particularly in the form of childcare for young children (Halim et al., 2023). These responsibilities can impede women’s labor supply and earnings. In Uganda, 37% of female business owners bring small children to work, compared with 0% of male business owners, and businesses where children are present earn 48% lower profits than female-owned businesses without a child present (Delecourt and Fitzpatrick, 2021).

Policy-makers increasingly recognize these challenges and are looking at childcare as a high-priority policy area to improve women’s labor supply and earnings while growing the economy. However, making childcare services available to low-income women entails large fiscal costs, as they have little ability to pay.² Given budget constraints, policymakers may wish to target support to women with the highest marginal returns to childcare. Workplace-based care offers a natural targeting mechanism: it provides the highest value to women who want to spend the most time at work, and it entails a small “hassle cost” that may deter women who realize few benefits from childcare from using it. By contrast, using existing privately provided, community-based childcare centers imposes fewer frictions in adoption for women and their families, potentially leading to larger economic gains.

We use a randomized control trial to test the impacts of two different models of childcare in a sample of working women across six markets in and near Kampala. The first intervention offered childcare services in “childcare corners” at the markets where women work. The second intervention provided women with access to existing childcare providers in the communities where they live. The two models had similar costs of provision but were fully subsidized from the perspective of the women in the study. Comparing both models to a control group of market women not offered free childcare speaks to the impact of childcare on

¹International Labor Organization, <https://ilostat.ilo.org/data/>.

²We find that women working in markets in Kampala are only willing to pay around one-fourth the cost of childcare services at baseline.

low-income women who are already attached to the labor force. Comparing the two models to one another illuminates key trade-offs in who takes up childcare and to whom the largest benefits accrue.

Pooling both treatments, we see that take-up of childcare increases by 48pp (229%). This leads to a local average treatment effect (LATE) of four fewer hours per day when the mother or a household member is the primary caretaker of the child, a 40% reduction. Families draw down external childcare support the most, virtually eliminating time when non-family members outside of the childcare centers are looking after the child. Women assigned the treatment group significantly reduce working hours, with LATEs showing 13% fewer working hours. However, earnings still rise, leading to more than 50% higher hourly earnings. Willingness to pay for childcare and our child development index also show strong, positive effects. Results are qualitatively larger among the poorest women in our sample, who use increased efficiency to expand earnings compared to wealthier women who use increased efficiency to reduce working hours.

Comparing the two models of childcare, we find that women offered market-based childcare are 41 percentage points (195%) more likely to be using formal childcare than women in the control group compared to a 55 percentage points (263%) increase for women assigned to community-based childcare. These differences confirm that market-based childcare carries implicit costs for many potential beneficiaries. In fact, we find that take-up of market-based childcare is lower than that of community-based childcare for every subgroup we examine. Accounting for these differences in take-up, LATEs show larger reductions in work hours for women in the market compared to the community treatment but larger increases in efficiency, leading to qualitatively larger increases in earnings. This is consistent with market-based childcare inducing women with the highest returns to select in. By contrast, women assigned to community-based childcare have significantly larger improvements in wellbeing, their children have larger increases in our child development index, and they respond with a greater increase in willingness to pay for childcare, consistent with community-based childcare offering meaningfully more convenience and potentially higher-quality childcare.

The contrast in uptake and impacts between the different models of childcare suggest meaningful policy tradeoffs. Community-based centers are, overall, more accessible and lead

to better women’s welfare and child development outcomes. By contrast, after accounting for differences in take-up, economic impacts are larger for women in the market-based group, although this difference between groups is imprecisely estimated. We use the framework proposed by [Hendren and Sprung-Keyser \(2020\)](#) to examine these trade-offs by calculating the marginal value of public funds (MVPF) for market- and community-based childcare. Our calculations show larger returns for subsidizing community-based childcare, which are driven by higher impacts on child development. However, if market-based centers could all be run at the per-child cost we estimate from the largest, most efficiently run market-based center in our study, market-based childcare would offer the better value proposition for government investment. Our analysis highlights that costs and gains to child development, which translate into higher earnings later in life, are the two most important factors in determining the value of publicly-supported childcare.

While there is a longstanding literature on childcare support in relatively wealthy countries, our findings contribute to a recently emerging literature focused on low-income settings where multiple cross-cutting constraints are likely, examining impacts of childcare among urban women in Kampala working in the informal sector.³ A number of recent studies have found that access to childcare in developing country settings can increase women’s labor force participation ([Martínez and Perticarà, 2017](#); [Clark et al., 2019](#); [Nandi et al., 2020](#); [Hojman and Boo, 2022](#); [Attanasio et al., 2022](#); [Halim et al., 2022](#)), but there is little evidence on changes in intensive-margin labor supply among women who are already working.⁴ We document that such women may not increase their labor supply when provided with childcare, but that they can still realize gains from working more efficiently. In addition, we find significant improvements in women’s well-being even in a population of women who had found a way to navigate working and childcare at baseline. Our findings help build the case for the value of childcare by documenting impacts even in the absence of extensive-margin entry into the labor force.

³Some notable examples of earlier studies of childcare impacts on women in wealthier settings include [Baker et al. \(2008\)](#); [Morrissey \(2017\)](#); [Heckman \(1974\)](#) with another set of papers focusing on child developmental gains from such programs [Havnes and Mogstad \(2011\)](#); [Duncan et al. \(2022\)](#).

⁴Also in Uganda, [Bjorvatn et al. \(2022\)](#) show the importance of credit constraints as a complementary constraint preventing returns to increased labor supply when women are provided with childcare support. In Egypt, [Caria et al. \(2022\)](#) find that strong social norms against the use of childcare support limited uptake of heavily subsidized childcare offers, impeding their ability to affect women’s income.

2 Background and Experimental Design

The study began in January 2023 with a listing of women working in six selected markets in the greater Kampala metro area.⁶ To be eligible women had to be i) at least eighteen years old, ii) the primary caregiver for at least one child aged eight months to four years, and iii) interested in receiving childcare support.⁷

Launching one market at a time, we interviewed eligible women who agreed to participate in the study in a baseline survey that took place between February and March 2023.⁸ After completing the baseline, we organized local public raffle events in each market to randomly and transparently assign women to a marketplace-based childcare treatment group, a community-based childcare treatment group, or a control group. Women in the two treatment groups then registered and enrolled in their respective childcare service centers to receive free childcare for 6–9 months.⁹ We conducted a follow-up survey with women near the end of these interventions, but while childcare services were still taking place. We ran a final, post-intervention, follow-up survey approximately three months after the end of the childcare interventions.

The study was designed to test the impacts of two different models of childcare support. These models were chosen to provide a contrast between different policy options being considered by Ugandan policy-makers as they seek to identify promising models of childcare to bring to scale: a public-styled marketplace-based childcare service and a private sector-led community-based childcare service.

The first form of childcare mimics a public investment approach. Many countries have been exploring ways to improve the infrastructure around public work spaces to be more inclusive of women. Frequently, this has included the expansion of childcare services offered at places of work with high presence of female workers, such as markets, where women

⁶The major considerations when selecting these markets were having a sufficient number of eligible women to justify the fixed costs of working in a given market, having space to establish the childcare corners within the markets, and covering a diversity of municipalities of Kampala.

⁷For children over three, we also required that the child was not already enrolled in formal school.

⁸Baseline data collection took place in this reference time for five of the six targeted markets. The sixth market started four months after the others.

⁹The shorter duration of childcare applied at certain markets where data collection was slower to start. It does not vary by treatment arm, as randomization occurred within markets.

comprise the majority of vendors and service providers. Although the market-based childcare centers in this study are not run by the government, we consider it a public investment *approach* in the sense that it involves providing a service in areas that private providers have not yet been willing or able to sustainably enter. We established the market-based centers by finding and leasing space in or near the markets. The research team helped to organize and establish these childcare centers, which did not exist at the time of the study's launch. We offered women in this group free access to these childcare centers for a period of 6-9 months.¹⁰ These market-based centers were only open to participants in the study who were assigned to that specific treatment group.

The second childcare model in this study leverages existing private providers in the communities where women live. We asked women in this group to identify private childcare centers of their choice with available capacity to include them. The research team then paid for the cost of enrollment for three terms (nine months), capped at a cost of 450,000 UGX (roughly USD 120) per term.¹¹ These pre-existing private childcare centers were open to anyone in the community willing and able to pay their child's tuition in addition to the study participants who were assigned to the community childcare group. These centers tended to be more explicitly focused on early child development credentials and activities than the marketplace-centers established as part of the study.

Table 1 summarizes these two forms of childcare using data from 58 of the participating private childcare service providers in the community group and five of the market-based centers.¹² On average, the market-based centers were smaller operations than the existing

¹⁰Selection of service providers who actually ran the marketplace-based childcare centers happened through a competitive process. A call for expression of interest was shared with members of the Early Childhood Development (ECD) Association of Uganda. After evaluation, we selected three service providers and allocated each one two centers to manage. The roles and responsibilities of the service providers included running of the day care facilities on a day-to-day basis, recruiting required staff and supervising staff, ensuring safety of children, creating a stimulating and child-centered environment, ensuring amicable relationships with the parents/guardians, and providing periodic reports on attendance and incidents at the center. To ensure quality and safety of these centers, a team from the Ministry of Gender Labor and Social Development (MGLSD) undertook bimonthly inspections at the centers and the research team conducted monthly monitoring visits.

¹¹If the center charged a cost greater than the cap, we allowed participants to register their children there if they wanted to pay for this balance themselves, but the project subsidy did not ever exceed 450,000 UGX. In practice this happened very infrequently but explains why average costs are slightly higher than 450,000 UGX in Table 1.

¹²One of the market centers was not fully operational at the time when we ran our center data collection.

private providers, occupying smaller physical spaces with smaller staff sizes, lower student to staff ratios, and slightly more availability per day (1.5–2 hours). The community-based centers had slightly more experienced teachers, on average, and bigger play spaces. Although not by design, the average per-child cost of running childcare centers in the market-based centers turned out to be roughly 499,207 UGX (USD 134) per term, very close to the median fees charged by the private providers. For the remainder of the paper we refer simply to the “market group” and the “community group” as shorthand for “marketplace-based treatment group” and “community-based treatment group” respectively.

The final study sample includes a total of 883 women across all six markets. We randomly assigned 254 women to the market-based childcare group, 257 to the community-based childcare group, and the remaining 372 to the control group.¹³ By randomly assigning women to the different treatment groups we can estimate the causal effect of access to these services and differences between the two forms of childcare.

3 Empirical Approach

3.1 Estimation

Our primary analysis estimates intent to treat effects of access to childcare services through either of the two interventions using an ANCOVA specification following McKenzie (2012):

$$outcome_i = \beta_0 + \beta_1 Childcare_i + \beta_2 outcome_{i1} + X_i + \psi_m + \epsilon, \quad (1)$$

where $outcome_i$ is one of a set of pre-specified outcome variables including women’s time use, labor outcomes, measures of women’s well-being, and child’s well-being for woman i . $Childcare_i$ indicates if woman i was randomly assigned to either of the treatment groups, meaning that β_1 is the coefficient capturing the effects of this access. We denote the lagged dependent variable $outcome_{i1}$ (where available). X_i are a set of individual controls chosen

¹³The original sample who participated in the baseline and raffle consisted of 972 women. Across all markets, targeted composition of treatment group assignments was 27.5% for each treatment group and 45% for the control group. The relative sizes of these groups were a function of budget and capacity constraints of the childcare centers. We held treatment group proportions fixed within each market (stratification unit).

using double-selection LASSO for precision, and ψ_m are a set of market (strata) fixed effects. We cluster standard errors at the individual level.¹⁴

When assessing differential impacts of the two childcare models, we estimate the following specification:

$$outcome_i = \alpha_0 + \alpha_1 Market_i + \alpha_2 Community_i + \alpha_3 outcome_{i1} + X_i + \psi_m + \epsilon. \quad (2)$$

Here, $Market_i$ indicates if individual, i , was assigned to the market-based childcare group and $Community_i$ indicates if that individual was assigned to the community-based childcare group. When examining differential impacts by subgroups, we interact treatment assignment with the relevant measure of heterogeneity along with the control variables.

Assessing the impacts of childcare among adopters, we use treatment assignment as an instrument for any reported usage of childcare in the preceding week.¹⁵ Both the intention to treat effects (ITTs) as well as the local average treatment effects (LATEs) can be informative. In thinking about value of impacts relative to cost, LATEs may better capture returns if policy-makers can trim out costs associated with participants who never take up or drop out of treatment. ITTs instead speak to average effects across the population that we study. We therefore preserve both throughout the analysis.

Our primary analysis focuses on six primary outcomes and indices: 1) time spent on childcare, 2) time spent on income-generating activities, 3) business performance, 4) mother's well-being, 5) child development, and 6) mothers' willingness to pay for childcare services. We report on supporting additional variables, including index sub-components, in the appendix.

3.2 Attrition and Balance

We conducted the study's primary follow-up survey six to nine months after the baseline. It had a high overall tracking success rate of 91%. Despite extensive efforts to track baseline

¹⁴For outcomes that were not captured during the baseline or for individuals missing that response at baseline, we fill in this missing baseline value with the mean of the control group at baseline along with a dummy indicator for missing that value, equal to one.

¹⁵For the pooled estimates we have two distinct instruments for childcare usage. When estimating separate LATEs, we follow [Erickson and Whited \(2002\)](#) in our main manuscript tables. Their approach allows us to estimate the LATE effects of each treatment separately within the same model and then conduct the t-test for significant differences between the two treatment groups.

4.1 Program Take-up and Primary Caregiver Responsibility

We begin by examining take-up of our two childcare interventions. In other settings, willingness to use childcare, even at heavily subsidized or free rates, has suggested that there may be meaningful non-monetary obstacles to providing women with childcare support. We do not observe such patterns in our setting. Consistent with women’s eagerness for childcare help, [Table 3](#) shows large and significant take-up for both treatment groups as reflected in their likelihood of having used childcare in the week before the survey.

Panel B shows that women in the market group are 41 percentage points (195%) more likely to use childcare than women in the control group. Women in the community group are 55 percentage points (263%) more likely to use childcare. Take-up is significantly larger for the community group, with women 23% more likely to have used childcare than women in the market group. This initial difference is consistent with community-based childcare being more convenient and accessible since it is located closer to women’s homes. Further, women in the community group were allowed to choose which center they wanted to use, which may reflect higher perceived valuation or convenience.²¹

Our results suggest that women who take-up childcare use it nearly every day when they are working. Women in the market group use it for an additional 1.6 days per week, on average, and those in the community group using it for 2.3 additional days per week. Accounting for differential levels of take-up between the two groups, our estimates imply that compliers in both treatment groups tend to use childcare for around 5 days per week. Given that most women in our sample work 5–6 days per week, it appears that women who take-up childcare shift a substantial portion of childcare to the childcare centers.

To understand how access to childcare changes caregiving responsibilities, we next look at how the reported primary caregiver for the target child changes with access to free childcare. The outcomes in [Table 4](#) capture the target child’s primary caregiver on the day before the survey. Panel A shows pooled ITT estimates of access to either form of childcare on primary caregiver. Column 1 is time the target child spent at day care, column 2 is time spent

²¹We discuss patterns of differential take-up and accessibility among relevant sub-groups in Section 4.3 which suggest that transaction frictions and non-monetary costs are likely driving the overall differences in adoption as opposed to differences in perceived quality or value of the different types of childcare.

for each of these domains. We show the sub-measures used to construct the indices and secondary, supporting outcomes in the appendix.

Table 5 shows ITT estimates in Panel A and LATE estimates in Panel B. Pooling effects from both treatments yields ITT estimates that represent an average across the two treatment groups relative to the control group. LATE estimates show averages across the two treatment groups *among compliers* who differ by treatment group. Most of our discussion emphasizes the LATEs, as we think this is the relevant measure for assessing benefits relative to operational costs, but we show both versions throughout.²²

First, we find only modest changes in women’s childcare responsibilities. Childcare access leads to a marginally significant increase in time women spend as a child’s primary caregiver, though this amounts to just 15 additional minutes per day among compliers and has a lower Lee bound that goes below zero. There is a moderately sized reduction in time spent on childcare as a secondary activity (multitasking) that constitutes a reduction of 19% among compliers in panel B, though the effect does not survive our multiple inference correction.

Next, we look at women’s labor supply, or the amount of time that women spend working on income-generating activities. Perhaps surprisingly, we observe a significant reduction in women’s time spent working by 1.13 hours (13%) in Panel B, though again this effect does not survive our multiple inference correction.²³ Despite this reduction in time spent working, our estimates qualitatively suggest that women’s earnings increase. In the LATE results in Panel B, the point estimate suggests an economically meaningful increase in earnings of nearly 20%, though our estimate is noisy. Putting these pieces together, we find that this reflects a 57% increase in work efficiency as captured by earnings per hour.²⁴ Our results suggest that the burden of childcare for women in our sample is not solely about time, but about the ability to reduce time multitasking and work more efficiently.

Although the policy arguments for expanding access to childcare frequently center on economic impacts, welfare effects for both the mother and her child merit close attention as

²²If costs were fixed before launching a childcare center, ITT impacts might be preferred in weighing costs and benefits. In the discussion section, we highlight that the majority of childcare provision costs are marginal, not fixed. Using LATEs for cost-benefit assessment implicitly considers it more likely that a service provider would reduce costs based on actual enrollment and usage.

²³**Table A2** shows other women’s primary time use categories, with the only significant change reflected in a small increase in transport time of roughly 15 minutes.

²⁴We show additional measures of economic performance in **Table A3**

well. We look next at two measures of welfare: an index of women’s well-being and an index of caregiver-reported child development for the target child. Our ITT estimates show a nearly 0.1 standard deviation increase in well-being while the LATEs suggest an increase of nearly 0.2 standard deviations, although both effects fall just short of statistical significance after multiple inference corrections. These results appear to be driven by improvements in women’s life satisfaction rather than increases in locus of control or reductions in depression and anxiety (see [Table A4](#)). We find larger and more pronounced impacts for child development, with childcare leading to a nearly 0.3 standard deviation increase in our index of child development among compliers, a result which survives multiple inference corrections at the 5% level of significance. Our result on child development is particularly important in allaying concerns of potential harms that could result from separating women from their children and instead points to the opposite: time spent in childcare advances early child development, potentially spurring long-term benefits beyond immediate the financial returns to the child’s mother.²⁵

Finally, in column 8 we find that women in our treatment groups increase their willingness to pay for a term of childcare by 25%. Experience with childcare increases women’s valuation of the service. However, although this constitutes a statistically and economically significant increase, it still implies that women’s willingness to pay for childcare is just one third of the actual cost of providing a term of childcare. This suggests that experience with childcare is not sufficient to close the gap between willingness to pay and market prices.

[Figure 1](#) summarizes impacts across our primary outcomes. Encouragingly, even while lacking precision for some outcomes, the direction of all impacts for women gaining access to childcare across our sample is positive.

Given limited public funds to support childcare access, understanding how to effectively target support is critical. We therefore examine how the impacts of access to free childcare differ by wealth as captured by women’s baseline levels of food security.²⁶ We summarize the results in [Figure 2](#). Less vulnerable women have a significant reduction in their time spent

²⁵We additionally show pre-registered secondary children’s outcomes in [Table A5](#).

²⁶To measure food insecurity we use the Household Food Insecurity Access Scale developed by USAID and FANTA ([Coates et al., 2007](#)). With the index based off of this measure we then split our sample at the median to assess differential impacts by baseline food security.

efficiency are similar whereas the imprecisely estimated point estimates for earnings is four times larger in the market than the community group. However, although gains in efficiency are naively significant, neither result survives multiple inference corrections. Turning next to our welfare measures, we see that the positive effects for women’s well-being as well as child development are both qualitatively stronger in the community group as well as more precise. These results align with better ease of use as well as modestly higher quality for the community centers. However, given that the marketplace centers were solely providing care and not following an early childhood development curriculum, we are encouraged by the positive point estimate on child development with an economically large point estimate of 0.2 standard deviations among adopters. Gains in both women’s wellbeing and child development align with a significantly more pronounced increase in willingness to pay for women in the community group.

Altogether, our results show qualitatively stronger economic responses for the market group than the community group, although it is important to note that our results on economic outcomes do not survive multiple inference corrections. For policy-makers intent on efficiently promoting economic outcomes, such differences suggest that the selection induced by the less accessible marketplace-based model may be effective at boosting women’s economic potential. However, community-based childcare shows stronger impacts on our measures of women’s wellbeing and child development. Preferences between these two models may ultimately depend on fiscal capacity and policy objectives.

4.4 Persistence of Effects: Three-month Follow-up

Finally, we examine whether any of our effects from temporary access to free childcare persist after the end of the interventions, using data from the post-intervention follow-up survey that happened three months after the market centers closed and the subsidies for community-based centers ended. **Figure 4** summarizes these results with the original during-intervention results plotted next to the impacts post-intervention. Broadly, we do not see any evidence of persistence in time use, economic outcomes, or women’s welfare. However, we find some persistence in the use of childcare among women in the community group, who remain ten percentage points more likely to use childcare than women in the control group.

gain in tax revenue, we assume that women and their children pay income tax at the lowest non-zero rate, which is 10% in Uganda. We use LATEs in all of our calculations under the logic that the government could structure a subsidy program such that only women who use childcare regularly are eligible. We provide further details on these MVPF calculations in [Appendix B](#).

Our calculations suggest that under a cost-sharing model where governments provide subsidies sufficient to close the gap between women’s WTP and the cost of centers. We calculate the market center MVPF at 2.8: for each dollar the government spends, it would generate USD 2.80 in social welfare (see [Table B1](#)). The MVPF for community-based childcare is 4.2, driven by larger estimated impacts on child development and, by extension, children’s future earnings. If, seeking to maximize access for a vulnerable population with limited willingness or ability to pay, the government instead fully subsidizes childcare, our estimated MVPFs are 2.3 for market-based childcare and 3.0 for community-based childcare. These estimates all compare favorably to MVPFs for pre-primary education interventions in low-income countries, which tend to be between 1.8 and 3.1 ([World Bank, 2022](#)).

We consider two optimistic cases to illustrate how returns vary under different scenarios. If the cost for market centers fell to USD 89 per term as considered in the previous section, our estimate of the cost in the largest, most efficient center in our study, then the MVPF for market-based childcare more than doubles to 6.5 under cost sharing and 3.9 under a full subsidy. It follows that government efficiency in its service delivery, running a public childcare program, could dramatically increase the value proposition. Alternatively, we assume that the impacts we estimate on child development would grow over time in line with the growth documented in [Grantham-McGregor et al. \(1991\)](#): an additional 40% in year 2 and 20% in year 3.³⁰ Under this scenario, the MVPF rises to 4 for market-based childcare and to 8.1 for community-based childcare under cost sharing. Returns to sustained childcare could be substantially higher if impacts on child development compound over time.

Finally, we recognize that our MVPFs are coming from relatively noisy estimates over short horizons. We therefore also look at how our MVPF calculations would change if we

³⁰[Grantham-McGregor et al. \(1991\)](#) document 40% higher impacts on child development after 2 years than after 1 year. We assume these would decay further in year 3.

gains in women's well-being, willingness to pay, and child development. Market-based care reaches fewer women but appears to have sorted in those with the highest economic returns, generating larger reductions in working hours and a larger, if imprecisely estimated, increase in earnings. The community model thus favors reach and welfare while the market model favors economic returns and targeting. The choice between them depends on which objective a policymaker weighs most heavily. The take-up patterns also point to practical targeting lessons. Uptake of market-based care depends heavily on where women live and whether they already bring their child to work, highlighting that results may differ markedly depending on transport options and norms about bringing children to work in different settings. Take-up of both models is higher among women with children under two, suggesting that mothers of the youngest children may be an effective group to target.

The study also surfaces a fundamental obstacle to sustainable provision in this setting. Women value childcare and their valuation rises with experience, but their willingness to pay remains well below the cost of providing it. Expanding access through full-cost pricing is therefore not a viable path for this population. Realizing the benefits we document will require substantial subsidies. This positions childcare less as a privately provided service and more as a form of social protection or investment in a public good, comparable to public primary schooling. Our estimates of the marginal value of public funds suggests that the returns to spending on childcare could be substantial, with each public dollar spent on childcare subsidies increasing social welfare by 4–7 dollars for market-based childcare or 4–8 dollars for community-based childcare. Our estimates compare favorably to other early childhood interventions in low-income settings and suggest that public spending on childcare could be a worthwhile use of funding. However, our estimates rely on strong improvements in child development. If childcare leads to increases in women's earnings but does not boost early childhood development, the welfare gains may be less than the cost of childcare provision.



Regular article

Community effects of electrification: Evidence from Burkina Faso's grid extension[☆]Maika Schmidt^a, Alexander Moradi^b,*^a Department of Economics, University of Sussex, UK^b Department of Economics, University of Bozen-Bolzano, Italy

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ABSTRACT

We study the effects of Burkina Faso's large scale electricity grid expansion 2008–2017, using both community and household-level data. We show that the timing of electrification was driven by engineering constraints and thus largely exogenous. We estimate the impact of electrification using a staggered difference-in-differences (DiD) approach, where not-yet treated communities serve as the comparison group. Despite low household connection rates, we find strong positive effects on luminosity, drinking water provision and school electrification, suggesting that grid connection enables community-level infrastructure. At the household level, we find increases in ownership of electric appliances and financial inclusion. Importantly, effects spill over to households that do not have an electricity connection.

1. Introduction

Universal access to affordable, reliable, and modern energy services is a key Sustainable Development Goal (SDG). While significant progress toward this goal has been made globally, sub-Saharan Africa lags behind, with nearly half its population lacking electricity as of 2018. The International Energy Agency (IEA) estimates that achieving universal electrification in the region by 2040 would require investments of around \$60 billion annually (IEA, 2019). Given these high costs, understanding the developmental impact of electrification is crucial.

In this paper we estimate the impact of community electrification in Burkina Faso on a wide range of development outcomes. Our analysis exploits the roll-out of electrification across 558 localities between 2008 and 2017.¹ We find that while more developed localities were prioritized for grid connection, the timing of electrification was driven primarily by engineering rather than economic constraints. This quasi-exogenous variation allows us to estimate electrification effects using a difference-in-differences (DiD) approach with staggered treatment,

comparing electrified to not-yet electrified communities. Additionally, we employ an instrumental variable strategy based on a theoretical “least-cost” grid network.

Our findings highlight substantial community-level benefits, despite relatively low household electricity uptake. Electrification led to a substantial rise in nighttime luminosity, more electrified schools and improved drinking water systems. We also find suggestive evidence of accelerated population growth in electrified localities. At the household level, electrification boosted ownership of electrical appliances and financial inclusion, even among households that lack private electricity access. These results hint at potential spillover effects, where benefits extend beyond direct users of electricity.

A vast body of literature documents positive effects of rural electrification on development, particularly outside of Africa. In the U.S., rural electrification boosted farm productivity and land values (Kitchens and Fishback, 2015), as well as agricultural employment and population growth (Lewis and Severnini, 2020). In Asia and Latin America, studies have found income gains, increased female labor force participation,

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¹ A locality refers to a settlement (town or village); the terms locality and community are used interchangeably.

and improvements in education and health (e.g., Barron and Torero, 2014; Grogan and Sadanand, 2013; Khandker et al., 2013, 2012; Lipscomb et al., 2013; Van de Walle et al., 2017). However, the evidence for sub-Saharan Africa is more mixed.² Several studies find limited income effects (Ankel-Peters et al., 2024; Bensch et al., 2011; Bernard and Torero, 2015; Dinkelman, 2011; Lenz et al., 2017) and uncertain firm productivity gains (Akpanjar and Kitchens, 2017; Peters et al., 2011). In fact, several studies questioned the cost-effectiveness of rural electrification programs in sub-Saharan Africa given the low uptake and high infrastructure costs (Ankel-Peters et al., 2024; Bos et al., 2018; Lee et al., 2020; Lenz et al., 2017).

Many benefits of electrification accrue at the community level, benefiting even households without direct electricity access, e.g. due to improved provision of public services, increased local economic activity, and spillovers from connected households. In fact, many alleged benefits work through this channel (World Bank, 2008; Chaplin et al., 2017). For the U.S. as well as Asian and Latin American countries, studies have documented substantial community-wide benefits of electrification (Kitchens and Fishback, 2015; Lewis, 2018; Lewis and Severnini, 2020; Chakravorty et al., 2016; Burlig and Preonas, 2024; Lipscomb et al., 2013; Khandker et al., 2013; Van de Walle et al., 2017).

Studies on community-level electrification impacts in the African context remain limited. Moreover, most research focuses on the *intensive margin* — the impact of increasing household connection rates within already electrified communities — rather than the *extensive margin*, which examines the effects of introducing electricity to previously unelectrified localities (Akpanjar and Kitchens, 2017; Bernard and Torero, 2015; Chaplin et al., 2017). Two notable exceptions are Dinkelman (2011), who studied labor market effects in newly electrified townships in post-Apartheid South Africa, and Lenz et al. (2017) and Ankel-Peters et al. (2024) who studied electricity uptake in Rwandan communities.

Spillover effects are likely more substantial at the extensive margin of electrification. When a community first gains grid access, it enables the provision of electricity-powered public goods, such as street lighting, water pumps, and improved health facilities, which benefit a broader population. Furthermore, if positive community effects are ignored, comparisons between households with and without electricity within the same community can lead to an underestimation of the overall impact.

Studies outside Africa focusing on the *extensive margin* of community electrification indeed confirm large positive spillover effects to non-electrified households (e.g. Khandker et al. (2013) for Vietnam and Van de Walle et al. (2017) for India). Preliminary evidence from Africa also points to externalities. In rural Kenyan communities, connected households share the benefits of electricity with neighbors, by letting them watch television or charge their mobile phones (Lee et al., 2020), while in Rwanda and Tanzania, electrified households contribute to neighborhood safety by providing external lighting (Bensch et al., 2011; Chaplin et al., 2017).

Thus, to fully grasp the importance of electricity, it is crucial to examine the provision of public goods and spillovers at the extensive margin. Our study offers new insights from Burkina Faso, focusing on both community and household levels. We find that electrification enables community-level infrastructure, which has some positive impacts on closely related outcome variables. This understanding is particularly important given the ongoing debate about the cost-effectiveness of grid expansion in sub-Saharan Africa.

The remainder of the paper is organized as follows. Section 2 describes the data sources. Section 3 outlines Burkina Faso's electrification strategy, while Section 4 details the estimation strategy. Results are presented in Section 5 and discussed in Section 6.

² African countries covered include Benin (Peters et al., 2011), Ethiopia (Bernard and Torero, 2015), Ghana (Akpanjar and Kitchens, 2017), Kenya (Lee et al., 2020), Rwanda (Bensch et al., 2011; Lenz et al., 2017; Peters and Sievert, 2016), South Africa (Dinkelman, 2011) and Tanzania (Chaplin et al., 2017).

2. Data

We compiled a geocoded data set on electrification roll-out in Burkina Faso from 2008 to 2017. To this we added data at the locality and household level. Details are explained in Web Appendix A.

Electrification. We have rich spatial information on the electrification strategy implemented in 2008, which effect we intend to study (Ministère des Mines, 2008, thereafter MEPRED). We created a list of all localities connected to the national grid and their years of electrification.³ The *Annuaire Statistique 2017* was our main source (INSD, 2018). With the help of a map showing the electricity grid in 2017, we geocoded the electrified localities as well as the power lines used to connect them. Overall, there were 660 localities which were connected to the grid between 1954 and 2017. For 20 localities, we were unable to match the locality name to a location. Out of 640 georeferenced localities, 558 localities were electrified between 2008 and 2017.

Luminosity. We use cloud free satellite imagery of stable night-light from (i) the Defense Meteorological Satellite Program (DMSP) and (ii) the Visible Infrared Imaging Radiometer Suite (VIIRS). DMSP measures luminosity in digital numbers (DN), for the years 1992–2013, ranging between 0 and 63 (capped), available at 30 arc-second grid resolution (ca. 1×1 km).⁴ VIIRS measures luminosity in radiance in nano Watts per square centimeter per steradian, for the years 2012–2018, at 15 arc-second grid resolution. VIIRS offers the higher quality recordings detecting light at lower spectra with higher spatial accuracy (Gibson et al., 2021).

We construct two measures of luminosity based on the highest pixel value lit within each locality's polygon.⁵ First, we define a measure of extensive margin, a dummy variable indicating whether the locality was visible from outer space. Second, we construct a measure of intensive margin to quantify the degree of luminosity for ever-lit localities. We merged the DMSP and VIIRS series into a single metric, adjusting the values of the former based on the statistical correlation of their luminosity readings during 2012 and 2013.⁶

Other Data at the Locality Level. Furthermore, we collected detailed information on population, health facilities, schools, roads, drinking water provision, and irrigation. The purpose of collecting this data is two-fold. First, these community characteristics allow us to assess the development status of localities prior to electrification, and thus let us study selection into electrification. Second, some of the measures are available over time and thus allow us to estimate the impact of electrification on community characteristics.

We have population counts from the 1996, 2006, and 2019 censuses (Institut National de la Statistique et de la Démographie, 2000, 2011; Ministère de l'Eau et de l'Assainissement, 2020b).⁷ We added population data for 2015 using the Burkina Faso - High Resolution Settlement Layer from Facebook and CIESIN (2019).⁸ We retrieved data

³ We do not have information on isolated mini-grids or photovoltaic systems. They are not considered part of the national grid.

⁴ Top-coding is not a concern in our context. For localities electrified after 2008, the highest DN is 42.

⁵ Alternatively, luminosity can be calculated using mean pixel values within a locality's polygon, or by creating buffers of different sizes (e.g., 1 2 or 5 km) around the centroid or the brightest pixel of the locality, in line with Min et al. (2013), Dugoua et al. (2018) and Burlig and Preonas (2024). Our results are robust to these alternative specifications.

⁶ Li et al. (2020) proposed combining the two series into a single "harmonized" dataset. However, in our context, this approach leaves a large discontinuity in 2014, coinciding with the transition between series (Web.Appx. Figure B.5). Furthermore, the harmonized data exhibits implausibly large heaping at DN = 6, strongly reducing annual variation. Thus, we opted to predict radiance from DN directly.

⁷ Census data often contain inconsistencies due to omitted, merged, re-named, or misspelled localities. We were unable to track 52 electrified localities across all censuses.

on health facilities in 2007 from [Ministère de la Santé \(2008\)](#). We have information on five types of facilities: (i) CHR: Centre hospitalier régional (N = 12); (ii) CM: Centre médical (N = 34); (iii) CMA: Centre médical avec antenne chirurgicale (N = 41); and (vi) CSPS: Centre de santé et de promotion sociale (N = 1355) and (v) dispensaries (N = 86).⁹ We geocoded annual census data of primary schools for the years 2003–2013 ([Ministère de l'Éducation Nationale et de l'Alphabétisation, 2019](#)). The source informs about electricity at school, student enrollment, school sanitation facilities and teaching materials. We retrieved the location of asphalt roads from Michelin road maps ([Jewab and Storeygard, 2020](#)). We collected data on the timing of AEPS/PEA drinking water installations, their power sources ([Ministère de l'Eau et de l'Assainissement, 2020a](#)) as well as installation year and number of manual water pumps ([Ministère de l'Eau et de l'Assainissement, 2018](#)). Finally, we digitized the location of irrigation systems in 2008 (N = 97) (Figure 5, [Ministère des Mines, 2008, 2008](#)) and major, permanent cattle markets from the 2019 Survey of Livestock Markets (Table 1, [Minot and Elahi, 2020](#)).

Household survey data. We took household and individual level data from the 1993, 2003, 2010, and 2014 rounds of the Burkina Faso DHS ([INSD and ICF](#)). For confidentiality reasons, DHS displaced the true GPS coordinates of urban and rural clusters by up to 2 and 5 km respectively ([Burgert et al., 2013](#)). In our setting, this built-in measurement error could lead to a potentially large attenuation bias, if DHS clusters are matched to the wrong localities. We therefore apply a novel technique to assign DHS household clusters to electrified localities. For each DHS cluster, we calculated a *probability score* that describes the probability of the cluster lying in a given locality. We then only included DHS clusters in our analysis with a probability score of at least 25%, excluding multiple matches with lower scores for the same DHS cluster. We show that this strategy leads to a sample representative of treatment times and administrative regions. Details of the matching strategy are described in Web Appx. A.3.

The 1993, 2003, 2010 rounds are standard DHS surveys, whereas the 2014 round is a Malaria Indicator Survey (MIS) offering less information on health and well-being. This means that we could only observe some development outcomes until 2010. To overcome this limitation, we added a survey from the Living Standards Measurement Study (LSMS) carried out in 2014.¹⁰ We identified LSMS cluster locations by region, province, commune and place name. Thus, we can perfectly match LSMS clusters to the correct electrified community and assign all of them a probability score of 1.¹¹ Overall, we have 219 DHS and 67 LSMS clusters, with roughly 22 households per cluster.

Summary statistics are reported in Web. Appx. Table B.2.

⁸ The settlement layer provides population estimates at a 1 arc-second resolution (approximately 30 m), based on satellite images and building recognition, to which then population derived from sub-national census data was proportionally allocated. We assigned population raster grids to localities following a procedure similar to [Baruah et al. \(2020\)](#) and [AFRICAPOLIS \(2022\)](#). First, we removed single hamlets from the layer. Second, we defined the boundary of a locality as the continuous built-up area within 100 m. Thus, we constructed a buffer of 100 m and joined all overlapping polygons. Finally, we calculated the total population within each polygon of a georeferenced electrified locality.

⁹ The CSPS are the first level of care corresponding to primary healthcare centers. Staffed with about three health workers and two unqualified volunteers, they are responsible for 5 to 23 villages, providing basic outpatient services, including maternity care and administering vaccination programs in the villages. CM and CMA are the first referral level and are equipped with beds and surgical facilities ([Marschall and Flessa, 2011](#)). CHR are regional hospitals.

¹⁰ This is the only LSMS survey available for our time frame of interest.

¹¹ Web Appx. Table A.1 report the number of DHS and LSMS clusters assigned to localities electrified from 2008 onward. We excluded the 1998 DHS survey round from our analysis. Only a few, selected places were sampled during this survey. As a consequence, the survey does not represent the electrified localities well.

3. Background: Electrification in Burkina Faso

Burkina Faso is a small, poor, overwhelmingly rural country in West Africa.¹² It is also one of the least electrified countries in the world.¹³ In 2004, the government announced a national electrification strategy aimed at reducing poverty and ensuring equitable access to electricity across provinces at the lowest possible cost. By 2008, concrete planning activities had advanced. “Mainstreaming Energy for Poverty Reduction and Economic Development” (MEPRED) developed the methodology and reviewed several supply options including (i) the rehabilitation and construction of diesel thermal plants, (ii) the development of photovoltaic solar energy, and (iii) the utilization of biomass. These options were ultimately deemed too expensive, which led Burkina Faso to prioritize the construction of a unified national grid and its integration with the grids of neighboring countries ([African Development Fund, 2009](#), hereafter ADF).

Fig. 1 illustrates the progress in electrification in Burkina Faso between 1993 and 2017. Prior to 2008, household electricity access increased gradually, but the number of electrified localities remained limited. Between 2008 and 2017, however, the number of electrified localities increased eightfold, from 81 to 660. Fig. 2 maps the expansion of the national grid during this period. Most of the large cities were electrified before 2008. Indeed, the capital Ouagadougou and the second-largest city gained access to electricity as early as 1954, and by 2000, 30 of the 32 towns with populations exceeding 10,000 in 1990 were connected. From 2008 to 2017, the grid extended to smaller, less populated localities. Despite this, the urban-rural divide remained stark. In 2014, 58.4% of urban households had access to electricity, compared to just 4.7% in rural areas.

In 2014, electricity supply accounted for approximately 1.3% of Burkina Faso's GDP. The state subsidized electricity. The subsidies amounted to 0.33% of GDP in 2016. Burkina Faso's electricity provider demonstrates slightly better efficiency than the average for African countries. In 2014, electricity losses, calculated as the difference between electricity fed into the grid and actually billed amounted to 17% (mean for sub-Saharan African countries: 22%, sd: 10) ([Trimble et al., 2016](#)). These losses include technical losses during transmission and non-technical losses such as theft, meter tampering, under-reading and under-recording of consumption. Furthermore, Burkina Faso's electricity reliability is on par with the average for African countries. According to the 2014 Afrobarometer (q94) survey, 68% of connected households reported having electricity “most or all of the time”, 17% “about half of the time”, and 14% “occasionally”.

Synergies of Electrification. The government's electrification plan proposed using electricity to strengthen the delivery of public goods such as (i) public lighting, (ii) lighting systems in schools, (iii) lighting and refrigeration for health centers, (iv) water pump systems, and (v) irrigation pump systems for off-season crops ([Ministère de l'Économie et du Développement, 2004](#); [Ministère des Mines, des Carrières et de l'Énergie, 2008](#)). In Section 5.2, we will examine whether outcomes can be attributed to electrification or were resulted from a broad bundle of coordinated development programs implemented concurrently.

Electrification was expected to have wide-ranging effects. [African Development Fund \(2009\)](#) speculated that semi-urban electrification would help the cottage industry¹⁴ and trading activities, engaged in

¹² In 2000, Burkina Faso had a population of 11.6 million, with a GDP/c (PPP) of 1300 (constant 2017 international \$) and 80% of the population employed in agriculture; in 2004, an estimated 57% lived below the \$1.90 a day poverty line ([World Bank, 2020](#)). Between 2000 and 2018, per capita income grew steadily by an average of 3.5% per year (sd: 0.6).

¹³ Burkina Faso's access to electricity ranked among the lower 20th to 30th percentile of sub-Saharan African countries in the 2000s and roughly maintaining this rank later on (Web Appx. Figure B.1).

¹⁴ The cottage industry refers to small-scale manufacturing activities carried out in homes.

mainly by women. In the health sector, electric refrigerators could replace unreliable gas-powered ones, improving the storage of medicines and vaccines. Furthermore, small medical devices and sterilization equipment could be used, and new health equipment installed. In education, electrification could support evening classes, enable internet-based learning, and improve working conditions for teachers and health personnel, ultimately improving attendance and retention rates (African Development Fund, 2009; Ministère des Mines, des Carrières et de l'Énergie, 2008). However, whether these anticipated effects materialized remains unclear. Even where grid electricity was introduced, photovoltaic systems may have continued to be the primary energy source for schools and health facilities (Ministère des Mines, des Carrières et de l'Énergie, 2008). Moreover, electricity may not have been the key constraint to development as assumed, but rather one of several interconnected challenges

Selection into Electrification. Ministère des Mines (2008) proposed extending the 33 kV network to localities with significant electricity demand and strong development potential. To identify these localities, they followed a systematic approach. First, they calculated a composite Development Potential Index (DPI) for each locality, incorporating three equally weighted dimensions: health, education, and economic development. Sub-indices included indicators such as the presence of health facilities, access to water, educational institutions, population size, agricultural facilities, markets, road infrastructure, and financial institutions (see Web Appx. Table B.1 for a detailed list and weights). Next, to maximize impact for the benefit of the greatest number of people, the population within the catchment areas of these localities was used to designate *development poles* that would be prioritized for electrification. In our analysis, we control for whether a locality is classified as a development pole, its DPI,¹⁵ as well as individual sub-indices.

Ministère des Mines, des Carrières et de l'Énergie (2007) deemed the most cost-effective electrification strategy to expand the network of 33 kV lines within a 50–60 km radius around the existing distribution lines. The 33 kV lines would typically serve the regional centers. Smaller localities could then be connected using single-wire earth return (SWER) lines. SWER lines are cheaper, because they use earth as the return path for the current eliminating the need for a second wire. However, their lower power transfer capacity makes them suitable only for areas with low electricity demand. Then, for localities farther away from the national grid, for which extending the grid would be prohibitively expensive, isolated off-grid diesel power plants were proposed. Finally, it was decided to extend the distribution network to nearby localities where geographic proximity to the new lines ensured economic viability.

Actual electrification certainly deviated from planning. Ministère des Mines (2008), for example, identified 329 development poles and proposed connecting 486 localities via 33 kV lines and 1037 via SWER by 2014. However, by 2017, only 527 localities had gained access to electricity. So what type of localities got electrified? Table 1 examines the covariates of electrification 2008–2017.

In col. (1), we find a significantly positive correlation between being assigned development pole status and the probability of getting actually connected to the grid between 2008 and 2017. In addition, a locality's own population and DPI are positively associated with electrification. These three variables are jointly highly significant ($\chi^2 = 806.7$, $p\text{-val} = 0.00$). Overall, more developed localities were more likely to become electrified.

In col. (2), we add additional measures of development not captured by the DPI, such as dummies for administrative capitals and the square

root of the number of primary school students. While coefficients for the DPI and pole decrease due to a certain extent of multicollinearity, the Pseudo R^2 increases. This suggests that the DPI does not fully capture the development factors influencing electrification. In other words, the decision to electrify a community depended on its development status, including factors beyond those measured by the DPI.

In col. (3), we introduce engineering considerations that determined the expansion dynamics of the grid. Specifically, we include a dummy variable for whether a locality was within 60 km of a 33kV line in 2007, the government's reported maximum distance for rural electrification at the time (Ministère des Mines, des Carrières et de l'Énergie, 2007). We also include the distance to the hypothetical 33 kV network that minimizes the line length when connecting all development poles (MST). We find localities further away from the pre-existing and optimal grid were less likely to be connected.

Selection in the Timing of Electrification. In col. (4)–(6) of Table 1 we use the same set of explanatory variables to test the covariates of the timing of electrification. Higher DPI localities got connected earlier (col. (4)). However, the correlation with development pole and population is rather weak. As before, adding development indicators not captured by DPI increases the Pseudo R^2 (col. (5)).

Interestingly, col. (6) suggests that engineering constraints played a large role in the timing of electrification. It turns out that the distance variables to the existing grid and the optimal grid (MST) are highly significant, whereas the DPI, development pole and population are jointly not significant ($p\text{-val}$: 0.13). Thus, the timing of electrification did not depend so much on a locality's development status. It appears that it is governed by engineering concerns, i.e. that the network has to expand from the existing network.

Overall, localities were selected for electrification based on their size, development and growth potential. The correlations highlight the serious endogeneity concerns when comparing electrified to non-electrified localities. However, we follow a different strategy comparing localities that were actually connected, but in different years. Those localities should be more similar, as they all qualified for electrification. Indeed, we found that the timing is far less correlated with development. The state of the 33 kV distribution network determined the timing when localities were electrified. Localities closer to the existing network were connected earlier. Moreover, it appears that there is a lot of noise in the timing - in fact, we do not find any correlation between planned and actual date of electrification ($p\text{-val}$: 0.55).¹⁶ Nevertheless, endogeneity issues may remain a concern, which we try to mitigate in three ways. First, we use observable, pre-existing indicators as controls. In particular, we control for the DPI score of each location. Second, we empirically test the assumption of conditional parallel pre-trends. Third, as an extension we exploit an instrumental variable strategy based on the engineering aspect of the electricity grid.

4. Estimation strategy

4.1. Difference in difference estimation with staggered treatment timing

We study the impact of electrification in Burkina Faso at two levels. First, using panel data on localities, we analyze community-level infrastructures and outcomes such as night-lights and electricity powered public services. This approach also allows us to assess whether electrification came as a bundle, with improvements in other public goods such as water, sanitation and education. Second, we match household data from multiple rounds of DHS and LSMS surveys to the electrified localities, using repeated cross-sections to analyze the effects of electrification at both the household and individual levels. Here, we can disentangle private benefits from community-wide impacts, including potential spillovers to non-electrified households within electrified localities.

¹⁵ Ministère des Mines (2008) published DPI scores for only a subset of localities. We reconstructed the DPIs for all localities using available sub-index data. Web Appx. Figure B.2 shows that our DPI closely matches reported DPIs, with regression coefficients matching sub-index weights.

¹⁶ The proposed date of electrification comes from Ministère des Mines (2008).

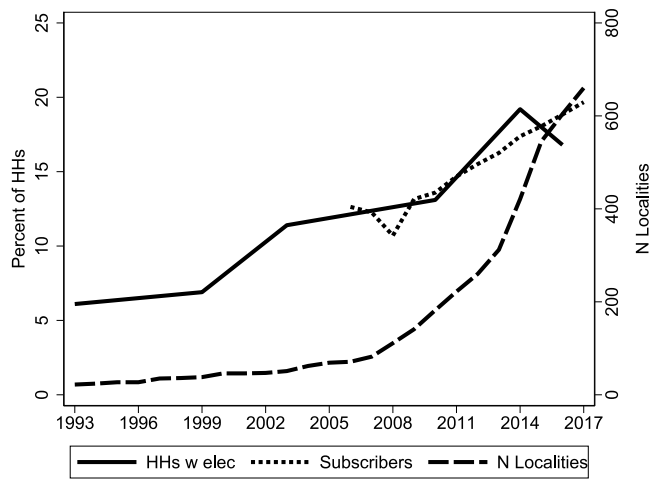


Fig. 1. Electrification in Burkina Faso 1993–2017.

Notes: This graph shows various measures of access to electricity. The percentage of households with electricity is from the 1993, 1999, 2003, 2010, 2014 and 2016 DHS (INSD and ICF, 2016). We also calculated the percentage of households that subscribed to the services of the electricity providers SONABEL and FDE. We divided the number of subscribers reported in Burkina Faso (2021) by the average household size derived from INSD and ICF (2016). Finally, the number of electrified localities (secondary axis) is sourced from INSD (2018).

Baseline estimation for the panel of localities. To capture the temporal dynamics of electrification, we employ a panel event-study approach. Using a panel of localities l observed over calendar years t , electrification acts as the *event*. Localities are treated in different years g_l , with $T_{lt} = t - g_l$ representing the years since electrification. Electrified localities are the *treatment group*, while those not yet electrified serve as the *comparison group*. Treatment is irreversible, meaning treated localities remain treated. In the absence of staggered treatment adoption or heterogeneous treatment effects, we could rely on the canonical two-way fixed effects (TWFE) model to estimate the dynamic impacts:

$$Y_{lt} = \alpha + \sum_{k=1}^K \gamma_k 1(T_{lt} = k) + \mu_l + \delta_t + X'_l \Gamma + \epsilon_{lt} \quad (1)$$

where Y_{lt} is the outcome variable, μ_l and δ_t are locality and year fixed effects, and X_l are a set of (time invariant) controls. The term $1(T_{lt} = k)$ is an indicator for the k th year relative to electrification.

However, staggered treatment adoption and heterogeneous effects may bias TWFE estimators (Borusyak et al., 2024; De Chaisemartin and D'Haultfoeuille, 2022; Sun and Abraham, 2021). To address these issues, we employ the imputation estimator proposed by Borusyak et al. (2024).¹⁷

The Borusyak et al. (2024) imputation estimator (called BJS from here on) proceeds in three steps. First, *within untreated observations only*, we estimate:

$$Y_{lt} = \mu_l + \delta_t + \epsilon_{lt} \quad (2)$$

Second, we use this model to predict the expected untreated potential outcomes for treated observations $\hat{Y}_{lt}(0)$. Treatment effects are then calculated for each treated observation as $\tau_{lt} = Y_{lt} - \hat{Y}_{lt}(0)$. Third, we aggregate these effects across observations. To estimate the treatment effects by length of exposure, we estimate τ_T , where $T_{lt} = t - g_l$ denotes the number of years since treatment. We also report the overall average treatment effect on the treated (ATT) τ , balanced with respect to event time.

Dynamic Estimation with Controls. To account for dynamic differences between localities with different development potential, we

extend equation (2) including a set of control variables that capture a locality's potential for future development. We interact these control variables with year dummies to allow flexible time trends based on observable pre-treatment differences.

$$Y_{lt} = \mu_l + \delta_t + X'_l \lambda_t + \epsilon_{lt} \quad (3)$$

where X_l is a vector of pre-treatment, time-invariant controls measured before 2008 and that were significant predictors of the year of electrification. Specifically, X_l includes whether the locality was designated a “development pole” by the central government, the Development Potential Index (DPI), and a measure of student enrollment in 2007. These pre-treatment measures of the community's development potential are then interacted with dummies for calendar years.

Baseline estimation for the repeated cross-sectional survey data. To estimate household and individual-level effects of community electrification using survey data, we use repeated cross-sections. Similar to the panel setup above, the classic TWFE estimator is unsuitable due to staggered treatment timing. We once again implement BJS. The regression model in the first step, estimated from only untreated observations, now becomes:

$$Y_{ilrt} = \alpha + \delta_t + \theta_r + X'_{lr} \lambda + A'_{it} \Gamma + \epsilon_{ilrt} \quad (4)$$

where i describes a household (or individual) observed at time t in locality l in region r . Y_{ilrt} is the outcome variable, and X_{lr} is a vector of control variables at the locality level (measured before 2008). A'_{it} is a vector of household level control variables that are time varying but exogenous to treatment (e.g., level of education among adult household members). Time fixed effects are denoted δ_t . In this set-up, individual (or locality) fixed effects cannot be included due to the lack of panel data. However, we include region fixed effects θ_r and we expand the set of control variables to counter the lack of locality fixed effects. Just like X_l in the locality-level estimation, X_{lr} includes whether the locality is a development pole, the DPI, and primary school enrollment in 2007. Additionally, X_{lr} includes whether the locality is a level 2 or 3 administrative capital, population before the treatment period (in natural logarithms), and survey fixed effects (whether the observation came from LSMS or DHS). By including X_{lr} we control for the covariates that were significant predictors of the year of electrification, as in col(5) in Table 1.

Dynamic estimation for the repeated cross-sectional survey data. To allow differences in trends based on a locality's development potential, we extend Eq. (4) by including community-level control variables interacted with year dummies:

$$Y_{ilrt} = \alpha + \delta_t + \theta_r + X'_{lr} \lambda_t + A'_{it} \Gamma + \epsilon_{ilrt} \quad (5)$$

allowing Y_{ilrt} to vary flexibly over time based on measures of the localities' development potential captured in the vector X_{lr} . These locality-level control variables in X_{lr} (the predictors of year of electrification from col (5) in Table 1) are measured before the first treatment period - i.e. pre 2008. We interact them with year dummies to allow flexible time trends based on pre-treatment locality characteristics. A'_{it} is again a vector of household level control variables that are measured at time t , but assumed to be independent of treatment.

Community versus private effects. The survey data includes information on whether households have private access to electricity. This allows us to disentangle the private effects of electrification from community or spillover effects. To estimate the latter, we re-estimate Eqs. (4) and (5) for households *without* private access to electricity. This yields a lower-bound estimate of the community- or spillover effects. This provides a lower-bound estimate, because it compares households in electrified communities that never connect to electricity (never-takers) with households in non-electrified communities, which include both never-takers and those that would connect if electrified (compliers). It is reasonable to assume that complier households differ substantially from never-taker households in their socio-economic status. To

¹⁷ We employ the Stata package `did_imputation` to implement the estimator.

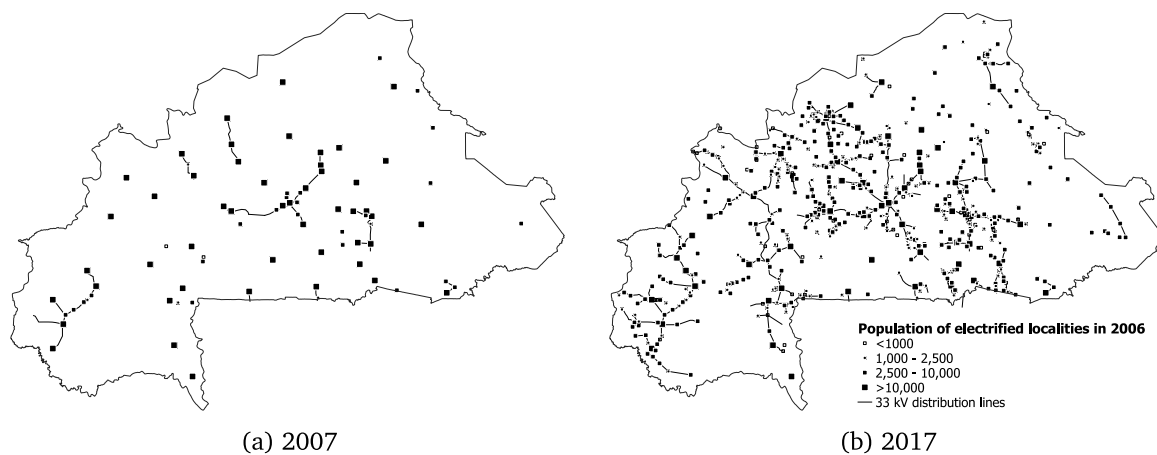


Fig. 2. Electrified localities 2007 and 2017.

Notes: The maps show the electrified localities and their population in 2006 (Institut National de la Statistique et de la Démographie, 2011). Fig. 2(a) shows the situation in 2007. N(localities <1000; 1000–2500; 2500–10,000; >10,000) = 2; 6; 26; 47. Fig. 2(b) shows the situation in 2017. N(localities) = 61; 184; 277; 56. The 33 kV power lines are the main distribution network. See Web Data Appendix for data sources.

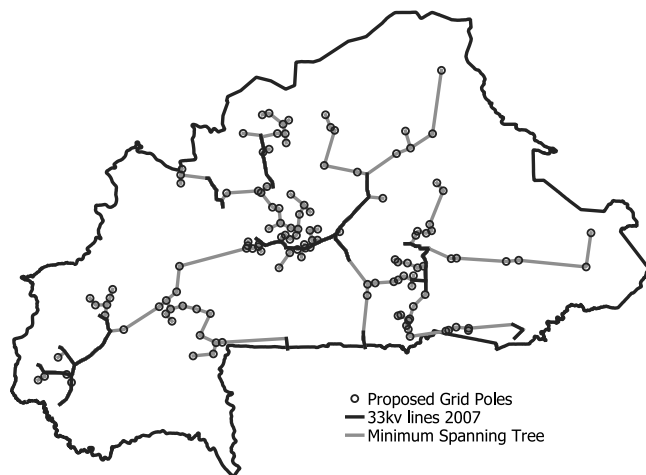


Fig. 3. Proposed grid poles and minimum spanning tree.

Notes: The map shows the proposed development poles by Ministère des Mines (2008). Taking into account the existing 33kV grid of 2007 the network that would connect the development poles minimizing network length is the Minimum Spanning Tree (MST).

address potential bias arising from these differences, we control for pre-determined socio-economic characteristics, including the gender and age of the household head and the primary and secondary education of adult women in the household.¹⁸

4.2. Identifying causal treatment effects

Identification. An obvious threat to identification is selection into electrification. Web Appendix Table B.2 indeed shows that at baseline electrified localities systematically differ from those not electrified. Further, Table 1 confirms that a locality's development potential was an important determinant of electrification. We argue that eligibility criteria were common to all electrified localities and that engineering constraints, particularly the state of the 33 kV distribution network, largely determined the timing of electrification. We address this selection bias when exclude never electrified localities and limit the

comparison group to those localities that were not-yet electrified at time t . However, this may not remove the endogeneity issue entirely.

First, towns that are electrified earlier may nevertheless differ in their trajectories from those that were electrified later. Thus, the parallel-trend assumption is key for identification. We initially assume (in the notation of the locality regression) that untreated potential outcomes follow a fixed-effects model: $E[Y_{it}(0)] = \alpha_i + \beta_t$. This assumes no systematic differences in trends across localities i in the absence of treatment. We can further weaken this assumption by allowing trends to vary based on time-invariant observables. Thus, the parallel trend assumption becomes: $E[Y_{it}(0)|X_i] = \alpha_i + X_i'\gamma_t$.¹⁹ We test the parallel-trend assumption using a standard Wald test for pre-treatment trends. However, as Roth (2022) has noted, this type of test frequently fails to detect minor violations of the parallel pre-trend assumption that can still meaningfully influence the estimated post-treatment impacts. Following Roth (2022), we calculate the worst-case pre-trend scenario consistent with our data, assessing the maximum potential violation of parallel trends that conventional tests might miss. Additionally, we evaluate the robustness of our findings by calculating the “breakdown” point of significant results due to minor changes in the slope of pre-trends (Rambachan and Roth, 2023).

Second, there must be no anticipation effects. We argue that this is very likely the case, due to the unpredictable nature of the exact arrival of the grid.

Finally, the estimation requires a sufficient coverage of untreated observations for the control-treatment comparison. Therefore, we cannot estimate effects for 2017 localities electrified in 2017, as they purely serve as control units. Naturally, for data that run only up to 2014, localities electrified 2015–2017 will serve as controls.

Internal versus external validity. Our analysis is based on localities electrified between 2008 and 2017. This strategy strengthens internal validity by addressing endogeneity concerns stemming from positive selection, as outlined below. However, excluding never-treated localities and focusing on this specific period may compromise external validity. If, for example, electrification were eventually extended to the remaining 7000 localities in Burkina Faso, and those electrified during the study period had greater potential for positive outcomes than the average locality, our estimated effects may overstate the impact for the population as a whole. This reflects an inherent trade-off between internal and external validity.

¹⁸ Data on men's education is unavailable in some surveys and is therefore excluded from the analysis.

¹⁹ The assumption for the household model with controls in levels is analogous: $E[Y_{it}(0)|X_i] = \alpha_{it} + X_i'\gamma_t$.

5. Results

5.1. Community-level effects (locality data)

We first examine whether electrification enabled community infrastructure development. Fig. 4 shows the evolution of luminosity, AEPS/PEA drinking water systems, and school electrification, while the upper panel of Table 2 reports the estimated Average Treatment Effect on the Treated (ATT) for $T = [0, 6]$ balanced with respect to event time, along with a conventional cluster-robust Wald test for pre-treatment trends $T = [-6, -1]$.

Electrification significantly increased luminosity. At $T = -1$, only 3.6% of localities had detectable night-lights. From $T = 1$ onward, this share rose sharply before stabilizing approximately four years post-electrification (Fig. 4(a)). The estimated ATT indicates a 39.5% percentage point (pp) increase in lit localities during the six-year post-electrification period. A similar upward trend is observed at the intensive margin (Fig. 4(b)).²⁰ This pattern suggests a temporary growth effect in luminosity leading to a sustained higher level of night-time illumination.

The literature typically does not distinguish between extensive and intensive margins. At $T = 4$, the combined effect on radiance is 0.3, equivalent to 0.8 DN.²¹ This increase is larger than the +0.5 DN observed in 141 Senegalese and Malian villages electrified between 1992 and 2009 (Min et al., 2013) and is twice as large as the +0.4 DN effect estimated for India (Burlig and Preonas, 2024). Increased luminosity may stem from various sources. Min et al. (2013), for example, attributed rising night-lights in Senegal and Mali primarily to street lighting, whereas much of the economics literature associates luminosity with broader economic activity (e.g., Henderson et al., 2012).

Furthermore, electrification benefited public infrastructures. AEPS (Adduction d'Eau Potable Simplifiée) are small-scale piped water systems with storage tanks and public standpipes sometimes serving several villages, whereas PEAs (Point d'Eau Autonome) are standalone water points, such as boreholes fitted with motorized pumps, providing localized access to safe drinking water. These systems often rely on electrical pumps, which, after electrification, can be powered by grid electricity instead of diesel or solar. The timing of new installations is striking. Grid-powered AEPS/PEA installations increase sharply starting one year after electrification (Fig. 4(c)). Most installations take place between three and five years post-electrification, with virtually no additions by year six. Since new installations add to the cumulative stock, by $T = 5$, 16% of localities had installed a grid-powered system.

Electrification led to a notable increase in the share of electrified schools. As Fig. 4(d) shows, the share of electrified schools rose immediately at $T = 0$, with an effect size of 12.7 pp for $T = [0, 3]$. This represents a substantial improvement from the pre-electrification baseline of 22% (Table 2).

Finally, we examine student enrollment and population growth. We find little evidence that electrification increased student enrollment (Table 2). In contrast, electrified localities appear to experience faster population growth than those not-yet electrified. The estimated 4 pp increase suggests a significant effect, likely driven by migration. However, this effect should be interpreted with caution, as it relies on population measurements from 2006 and 2015. In Section 5.4 we apply a TWFE specification that better suits the available demographic data.

Results are qualitatively similar across models with and without controls (Table 2). As for the parallel-trend assumption, we cannot reject the null that pre-treatment coefficients are jointly zero in either

of the two models. To further validate the parallel-trend assumption, we conducted a power analysis following Roth (2022). Results, shown in Web Appendix Figure B.7, indicate that point estimates are well above the continuation of the worst detectable pre-trend consistent with our data. However, results for water provision and school electrification display greater sensitivity; the ATT loses significance if pre-trend slopes are adjusted upward by about 25% (Web Appendix Figure B.8).

5.2. Electrification or a bundle of development programs?

Electrification can improve the delivery of other public services. Consequently, policymakers may seek to pair electrification with other development initiatives to amplify the overall developmental impact. This raises an important question: Are the observed effects driven by electrification, or are they the result of accompanying initiatives? To explore this, we focus on education and water.

Education. We examine additional indicators of school infrastructure that are typically targeted by development programs in education (UNESCO, 2022). If school investments were part of a coordinated effort alongside electrification, we would expect to observe improvements in school infrastructures unrelated to electrification. However, we find no significant effects, either pre- or post-treatment, across a wide range of schooling investment dimensions, including the number of public primary schools, school sanitation facilities, and teaching resources (Panel B, Table 2).²² Overall, the evidence suggests that while grid expansion was used to electrify schools, it did not coincide with broader investments in school infrastructure or education.

Access to Clean Water. Strategic planning documents emphasized synergies between electricity and water development (Ministère des Mines, des Carrières et de l'Énergie, 2008; CIFAME (Commission Intersectorielle de Facilitation de l'Approche Multisectorielle dans le domaine de l'Énergie), 2008). Was electrification bundled with water investments?

First, the decentralized nature of water provision in Burkina Faso makes systematic coordination unlikely. In peri-urban localities, water supply is managed at the municipal level, often through Water Users Associations (Humphreys and Schwartz, 2018; Ouedraogo, 2010). The state-run utility, ONEA, primarily serves large urban areas. Indeed, ONEA played a negligible role in our sample, managing water systems in only five localities (Ministère de l'Eau et de l'Assainissement, 2020a). Moreover, municipalities receive minimal central funding: between 2007 and 2015, between 2007 and 2015, direct transfers to municipalities accounted for merely 8% of public expenditure on water supply (Humphreys and Schwartz, 2018). Instead, municipalities heavily rely on donor funding and service fees. The multitude of donors and the fragmented governance structure complicate coordinated water provision. For instance, Humphreys and Schwartz (2018) report 16 different donors providing funding for water projects across just four municipalities.

Second, we employ a placebo strategy to test for bundling of electrification with water infrastructure investments. If such bundling existed, systematic patterns in water provision following grid expansion would be expected. However, our data does not support this. Human-powered pumps continued to be widely installed post-electrification; between 2008 and 2016, localities already connected to the grid installed 1274 human-powered water pumps. While these manual pumps may complement grid-powered systems in areas unsuitable for electrical infrastructure, a coordinated rollout would imply a distinct increase synchronized with grid expansion. However, row 1 of Panel C Table 2 indicates that the installation of human-powered pumps neither surged nor declined systematically with electrification.

²⁰ Web Appx. Figure B.4 reports separate estimates for DMSP and VIIRS. Conclusions remain unchanged.

²¹ This conversion is based on the 2012/13 correlation between DMSP and VIIRS, where one radiance unit in VIIRS corresponds to 2.7 DN in DMSP.

²² Wall maps were selected as a representative indicator due to their highest correlation with the first principal component in a principal component analysis. Results for the available set of teaching materials are provided in Web Appx. Table B.3.

A similar logic applies to substitutes for grid power. Simplified Drinking Water Supply Systems (AEPS) and autonomous water access points (PEA) can be powered by any energy source. In the presence of coordinated bundling, we would expect either a substitution of grid power for solar power or an increase in solar installations coinciding with grid expansion if they were complementary. Again, solar remained an important primary energy source for electric pumps used in AEPS/PEA water systems, but the data indicates no systematic pattern: Solar-powered water system installations show neither significant pre-trends nor notable changes post-electrification.²³

Taken together, grid electricity is unlikely to have been well-coordinated with other development programs. Instead, grid expansion appears to have enabled water provision by reducing costs, rather than being part of a systematically bundled development initiative.

5.3. Community versus private effects (survey data)

Fig. 5 shows the impact of community electrification on household electricity uptake and asset ownership over time. Visually, there are no apparent conditional pre-trends discernible (a full discussion of the diagnostics for parallel trends follows below). The analysis encompasses two specifications: the first controls only for level-differences in pre-treatment community characteristics. The second specification allows for time-trends to vary flexibly by pre-treatment community characteristics.²⁴ Results are very similar across the two specifications. Household electricity access is virtually zero before electrification but then increases to around 20% six years after grid connection. We also observe small increases in ownership of radios, televisions, and refrigerators. Estimates for individual event-times are noisy, as the composition of treatment and control locations varies across survey rounds. For example, confidence intervals tend to be very large from $T = 5$ onward, because our sample only extends to year $t = 2014$, meaning we can only observe $T = 5$ for localities electrified in 2008 or 2009.

Due to the noise in individual event-time estimates, we estimate the simple ATT for 0 to 6 years post-electrification (Table 3). Just as in Fig. 5, we present results for the two alternative specifications (columns entitled “Levels” and “Trends”). For household assets other than electricity, we estimate both the *total effects* of community electrification, which include all households, and the *spillover effects* on non-electrified households. Electricity uptake is significant but remains low, reaching about 10% of households from a near-zero baseline. We find large and significant increases in radio ownership of 6 to 10 percentage points, with spillover effects identical to the total effect. In contrast, while community electrification significantly increases television and refrigerator ownership, these effects are limited to electrified households, likely because these appliances require a continuous power supply.

We also estimate the effect of community electrification on household access to piped drinking water, whether through private connections or public standpipes. Before electrification, only 5% of households had access to piped drinking water, a share that nearly doubles post-electrification. This finding aligns with the observed expansion of water provision systems at the community level. While spillover effects are imprecisely estimated, point estimates suggest they are only slightly smaller than the total effect.

The household level effects of community electrification accrue slowly over a number of years, as can be seen in Fig. 5. To take this into account, we separately estimate short- and medium-term effects of community electrification on household electrification and asset ownership. The results are shown in Web Appendix Table B.4. Immediate effects, 0

to 1 years after electrification, are very limited. However, medium-term effects 2 to 6 years after electrification are larger and more statistically significant than the overall ATTs reported in Table 3. Electricity uptake over this medium-term is on average 15 percent, and access to piped water increases by 7 percentage points, an impressive 140% increase from pre-electrification levels.

We report the p -value from a Wald test of joint significance of pre-treatment coefficients (6 years before electrification), none of which indicate statistical significance for any dependent variable. To assess whether pre-trend tests may be underpowered, we apply the diagnostics from Roth (2022) and Rambachan and Roth (2023) (Web Appendix Figures B.9 and B.10). The results on household electricity access are not sensitive to potential violations of the parallel trends assumption, while for the other variables there is a risk that pre-trend tests are underpowered. This is likely due to small sample size when estimating coefficients for individual time periods.

Finally, we tested but did not find any significant effect of community electrification on child health outcomes like infant mortality rates and child malnutrition (stunting and wasting), or health inputs such as prenatal care and birth attendance (Web Appendix Table B.5).

5.4. Alternative estimation strategies

The dynamic staggered regression estimator is our preferred specification, because it nicely illustrates the dynamics and makes efficient use of sample size. In this section we apply alternative identification strategies. First, we apply a standard static two-way fixed effects estimator (TWFE) to a reduced sample of villages, such that there are only two periods and two groups. Reducing our sample to a two by two matrix means that a standard TWFE estimator will be unbiased, but it requires a large reduction in sample size and change in sample composition. Second, we employ an instrumental variable approach to a cross-section of villages. This serves as a robustness check and additionally allows us to analyze additional development outcomes for which we do not have long panel data.

Canonical DiD setup. We follow a standard DiD set-up and estimate a static two-way fixed effects estimator (TWFE), as in Eq. (1), but with only two periods and two groups. For outcomes at the level of locality (l) in region r :

$$Y_{lrig} = \alpha + \beta \text{Treat}_g * \text{Post}_t + \mu \text{Treat}_g + \delta \text{Post}_t + \theta_r + X'_{lr} \Gamma + \epsilon_{lrig} \quad (6)$$

where g is now divided into a treatment group with $\text{Treat}_g = 1$ if $g = [2008, 2009]$ (localities electrified in 2008–2009) and a comparison group with $\text{Treat}_g = 0$ if $g = [2015, 2017]$ (localities electrified in 2015–2017). We observe both groups in a pre-treatment period ($\text{Post}_t = 0$ if $t = [2003, 2007]$) and in a post-treatment period ($\text{Post}_t = 1$ if $t = [2010, 2014]$). δ captures differences between the pre- and post periods. μ captures time-invariant differences between treatment and comparison group. β is the DiD estimate. For household-level outcomes the model adds household controls, just as in the main specification. Standard errors are clustered at the locality level. For the locality level analysis, we have yearly data for the pre- and post-treatment periods, whereas for the household level analysis the pre-period consists of the 1993 and 2003 DHS surveys, and the post-period consists of the DHS and LSMS surveys from 2010 and 2014.

Instrumental Variable Approach. We previously showed that conditional pre-trends do not statistically differ between localities electrified in different years, indicating that the identifying assumption may plausibly hold. However, some variables exhibit low statistical power in the pre-trend tests. As a robustness check, we employ an alternative identification strategy using an instrumental variable approach.

The timing of grid expansion is largely dictated by engineering constraints. Grid extension follows a contiguous process — standalone grids are of little practical use. Thus, localities closer to the existing network tend to be connected earlier, while those positioned along expanding transmission lines were connected as the network extended.

²³ Web Appendix Figure B.6 plots the pre- and post-electrification coefficients.

²⁴ In both models we also control for household characteristics, which, while not affecting point estimates, improves precision.

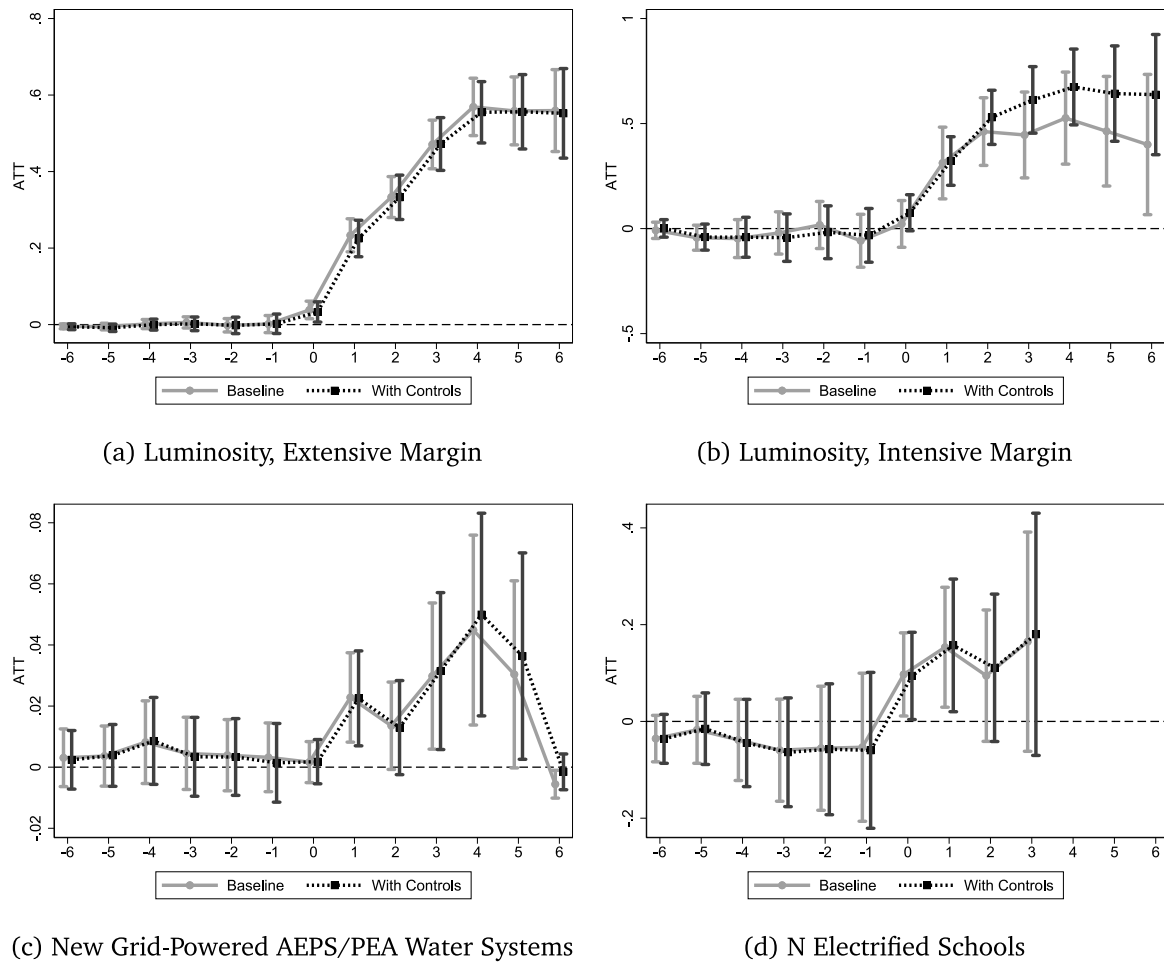


Fig. 4. Luminosity, water supply, and school infrastructure before/after electrification.

Notes: This figure presents average treatment effects on the treated (ATT) by length of exposure to electrification, along with 95% confidence intervals. The horizontal axis represents the length of exposure to the treatment ($T = [-6, 6]$). The zero line refers to the average of all pre-treatment estimates. The estimator is the imputation estimator proposed by Borusyak et al. (2024), and standard errors are clustered at the locality level. Pre-treatment parameters are reported as short differences ($Y_t - Y_{t-1}$). Sub Fig. 4(a) depicts the proportion of localities with non-zero night-lights (extensive margin), while Sub Fig. 4(b) displays the maximum radiance within ever-lit localities (intensive margin). The DMSP and VIIRS night-light datasets were harmonized using the correlation between the two series during their overlapping years (2012/2013). Sub Fig. 4(c) shows newly installed water systems (AEPS/PEA) powered by grid electricity. See Web Data Appendix for data sources.

However, using distance to the existing network as an IV would be naïve, as the existing grid is likely correlated with economic activity, with more remote localities being systematically less developed. Column (4) of Table 1 confirms this pattern.

To construct a valid instrument, we model a hypothetical least-cost grid network. We assume that the government wants to electrify all localities designated as “development poles” in the cheapest way possible. Taking the 2007 existing grid network and development poles as vertices, we construct a minimum spanning tree (MST) that minimizes total network length. Since network length is the primary cost determinant, this MST approximates the least-cost network. The resulting network is shown in Fig. 3. The log distance of the (non-development pole) communities to this least-cost electricity network then serves as an instrument for the timing of electrification. The instrument effectively captures complier localities that were connected earlier because they happened to be “on the way” as the grid was expanded to the development poles.

The key identifying assumption behind this instrumental variable strategy is that a locality’s distance to the least-cost electricity network affects the timing of its electrification but does not directly influence economic outcomes beyond its effect on grid access. To address potential endogeneity from variables used in the modeling of the least-cost network, we condition for proximity to the 2007 grid (within 60 km) and log distances to the development poles.

Constructing hypothetical - and therefore exogenous - networks is a well-established strategy in the infrastructure literature. Faber (2014), for example, constructed a hypothetical least-cost spanning tree network to model the expansion of China’s National Trunk Highway System. Similarly, Lipscomb et al. (2013) simulated the expansion of Brazil’s electricity grid 1960–2000 under purely geography-based cost considerations. Other studies emphasize network connectivity as the primary determinant of expansion. Jedwab and Moradi (2016) modeled a hypothetical least-cost colonial African railway network, predicting how major cities would have been connected under a unified system. Frye (2024) constructed an IV from a proposed but never-built historic highway plan, applying an algorithm that prioritized segments based on their importance for network connectivity.

Our IV set-up is then as follows. At the locality level, outcomes are measured between 2010 and 2014 and treated as a pooled cross-section. Localities electrified in 2008–2009 are the treatment group. Localities electrified in 2015–2017 represent the comparison group (as they were not yet treated in 2010–2014). Our instrument is log distance to the MST. We simultaneously control for whether the locality is a pole and distance to the nearest pole to take into account that economic activities may be spaced concentrically around poles. Complifiers are then localities that were connected earlier because they were located “along the optimal grid network”. We estimate the reduced form (RF)

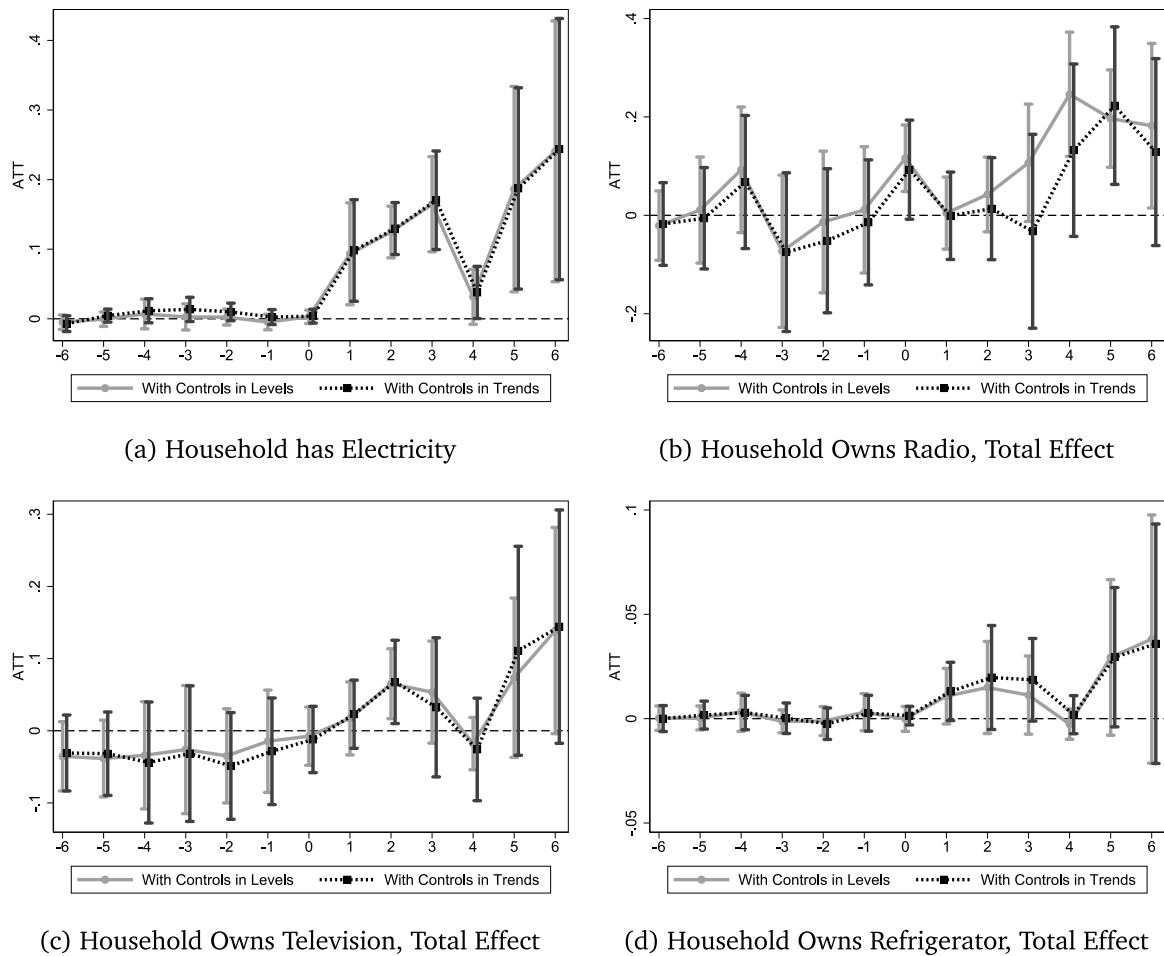


Fig. 5. Household access to electricity and electric appliances before/after electrification.

Notes: This Figure shows average treatment effects by length of exposure to electrification with 95% confidence intervals. The horizontal axis indexes the length of exposure to the treatment $T = [-6, 6]$. Estimator is the Borusyak et al. (2024) imputation estimator and standard errors are clustered at the locality level.

and the 2SLS. For outcomes at the locality level, the reduced form can be written as

$$Y_{lrt} = \alpha + \beta \ln(\text{Distance to MST})_{lr} + X'_{lrt} \gamma + \theta_r + \epsilon_{lrt} \quad (7)$$

for locality l in region r and years $t = [2010, 2014]$. Control variables X include indicator variables for the locality being a development pole, 60 km to the nearest 33 kV line in 2007, province and commune capital, and number of primary school students in 2007, logged distance to pole and the DPI. For outcomes at the household level, the model additionally includes household-level controls, and standard errors are clustered at the locality level.²⁵

Note that the IV generates a LATE estimate (not an ATT). In addition, the sample has changed, because we dropped localities electrified in the years $t = [2010, 2014]$. Hence, results may not be directly comparable to the BJS estimates from Tables 2 and 3.

Results for the community-level variables are presented in Table 4. Starting with the ATT using the BJS imputation method for localities electrified between 2008/2009 and 2015–2017, we find significant increases in luminosity at the extensive and intensive margin, AEPS/PEA drinking water systems, and electrified schools. The canonical 2×2

DiD estimator, while producing slightly larger estimates for luminosity and smaller for school electrification and water systems, supports these findings.

In subsequent columns, we restrict our analysis to the post-treatment period 2010–2014. OLS essentially estimate the difference between electrified and non-electrified localities conditional on controls. The estimated effects lie in between the BJS and TWFE point estimates.

Finally, the instrumental variable regression produces point estimates for the local average treatment effect (LATE), which tend to be higher than the baseline OLS. Notably, the effect on nightlight intensity triples while the electrification of schools doubles, while the effect on student enrollment becomes positive. However, the estimates are not statistically different from the BJS, TWFE or OLS estimator. As indicated by the Kleibergen–Paap and Montiel–Pflueger F-statistic, the instrument strength is adequate, except for luminosity at the intensive margin, where the large drop of non-lit localities weakens the instrument. Overall, this battery of alternative identification strategies corroborate the robustness of our findings from the staggered DiD model, pointing to consistent improvements in community infrastructure post-electrification.

Results at the household level are reported in Table 5. The first column shows our preferred dynamic estimator, as used in Table 3, but applied to the reduced sample. The second column shows the standard TWFE estimator (also applied to the reduced sample). The two DiD specifications yield very similar estimates of the ATT. Both show a large and significant increase in household electricity uptake, as well as significant increases in television and refrigerator ownership. These

²⁵ Community control variables are the same as in the main specification, but we have added the logged distance to the nearest development pole and whether the locality is within 60 kms of the nearest 33 kv line in 2007. These distance measures allow us to isolate the exogenous part of the variation in our instrument: the logged distance to the MST.

Table 1
Covariates of electrification 2008–2017.

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. var./Estimator	Electrified 2008–2017/Probit			Year of Electr./Ordered probit		
Sample	Localities not yet electrified in 2007			Only localities electrified 2008–2017		
Development Pole	0.81*** (0.07)	0.48*** (0.09)	0.52*** (0.09)	0.08 (0.11)	0.24** (0.12)	0.28** (0.13)
LN(Population)	0.30*** (0.04)	0.28*** (0.05)	0.32*** (0.05)	0.07 (0.08)	0.13* (0.07)	0.08 (0.09)
DPI	4.44*** (0.39)	2.62*** (0.47)	2.03 (0.48)	−2.56*** (0.56)	−1.48 (0.76)	−0.73 (0.63)
Within 60 km to grid 2007			0.09 (0.10)			−0.49*** (0.16)
LN(Dist. to grid 2007 (km))			−0.12** (0.04)			0.07 (0.06)
LN(Dist. to dev. pole (km))			0.12** (0.06)			−0.06 (0.09)
LN(Dist. to MST (km))			−0.20*** (0.05)			0.26*** (0.09)
Observations	7672	7672	7332	502	502	502
Region FE	YES	YES	YES	YES	YES	YES
Dev. Indicators	NO	YES	YES	NO	YES	YES
Pseudo R2	0.290	0.322	0.338	0.044	0.052	0.080

Notes: This table tests covariates of electrification. Unit of observation are localities. Localities electrified before 2008 are excluded. Col. (1)–(3) estimate a probit model where the dependent variable is a binary indicator of whether the locality was electrified between 2008 and 2017. Col. (4)–(6) estimate an ordered probit model where the dependent variable is the year of electrification, treated as an ordinal outcome. Development pole is a dummy indicating whether the locality was identified as such by *Ministère des Mines* (2008) due to a high DPI value and population in the catchment area. The Development Potential Index (DPI) is a composite measure of development indicators (see Web Appx. Table B.1 and estimated for all localities using available data. The Minimum Spanning Tree (MST) is calculated using the proposed 33kV poles. “Dev. Indicators” refers to additional indicators including the square root of the number of primary school students and dummy variables for administrative capitals (levels 2 & 3). All regressions include region fixed effects. Standard errors in parentheses.

*** p<0.01, ** p<0.05, * p<0.1.

Table 2
Average treatment effects of electrification.

	Mean (T = -1)	Baseline		Controls		Pre-Trends (<i>p</i> -value)		Nt/N
		ATT	se	ATT	se	Baseline	Controls	
Panel A: Outcome variables								
1. Luminosity, Ext. Margin	0.04	0.395***	(0.026)	0.394***	(0.029)	0.320	0.145	1842/13 329
2. Luminosity, Int. Margin	0.02	0.376***	(0.089)	0.498***	(0.071)	0.551	0.527	1124/6766
3. New grid-Powered AEPS/PEA water systems	0.00	0.020***	(0.005)	0.023***	(0.005)	0.939	0.911	1842/17 135
4. N electrified schools †	0.22	0.127**	(0.064)	0.139**	(0.071)	0.252	0.203	440/5353
5. N students in public primary schools †	52.08	-3.482	(2.809)	-0.054	(3.273)	0.171	0.382	440/5353
6. Annualized population growth (%)	-1.88	4.269	(3.522)	4.261	(3.542)	0.657	0.387	366/1899
Panel B: Testing for simultaneous schooling investments								
1. N public primary schools †	0.75	-0.029	(0.036)	0.003	(0.042)	0.419	0.657	440/5353
2. N latrines †	1.32	0.149**	(0.071)	0.012	(0.076)	0.746	0.397	440/5353
3. N wall Maps †	3.37	0.140	(0.199)	0.218	(0.218)	0.168	0.265	440/5353
Panel C: Testing for coordinated water investments								
1. New manual water pumps	0.45	-0.016	(0.042)	0.003	(0.323)	0.323	0.316	1842/17 135
2. New solar-Powered AEPS/PEA water systems	0.02	-0.011	(0.009)	-0.006	(0.320)	0.320	0.142	1842/17 135

Notes: This table reports Average Treatment Effects on the Treated (ATT), balanced with respect to event-time for $T = [0, 6]$, and for $T = [0, 3]$ for variables marked with a dagger †), using the imputation estimator of *Borusyak et al.* (2024). Panel A presents key outcome variables, while Panel B and C cover broader investments in school and water infrastructure. For each variable, we estimate two models: i) a “Baseline” model with locality and year fixed effects; and ii) a “Controls” model adding development-pole-by-year fixed effects and time-invariant controls (as measured in 2007) interacted with year dummies. These controls include the square root of primary school enrollment in 2007 and the predicted development potential index. In Panel B, the number of wall maps serves as a representative indicator for teaching materials. Results for additional teaching materials are provided in Web Appx. Table B.3. Panel C reports new installations of hand- and foot-operated pumps as indicated by the pump trade marks India, India Mark II, Vergnet, ABI, DIAFA, Kardia, and Volanta. The test for parallel pre-trends is conducted by regressing the outcome on dummies for periods $T = -6, \dots, -1$ using only non-treated observations. This corresponds to a cluster-robust Wald test, assessing whether pre-treatment trends are statistically different from zero. The final column reports the number of treated units (Nt) across all horizons and the total number of observations (N) used in the estimation.

effects are larger than in the full sample (Table 3); this could be either due to effect heterogeneity across localities, or due to the different time structure in this reduced sample, where we observe more localities longer after electrification. The coefficients for radio ownership and access to piped water are positive and of similar magnitudes as in the main estimation, but they are not statistically significant. The IV estimates show very large and significant effects for electricity access,

radio, and television ownership. The Montiel–Pflueger F-statistics are reasonably large, suggesting that the instrument is not weak.

An important advantage of the IV strategy is that it only requires cross-sectional data, allowing us to expand the analysis to data that were not collected in the earlier survey rounds. Table 6 shows results for additional household level variables available only for 2010 and/or 2014. Just as before, localities electrified in 2008–2009 are

Table 3
Household asset ownership - Average treatment effects (0–6 years after electrification).

Dependent variable	Total effect		Spillover effect		Pre-trend p-value		Mean Dep t=−1	N
	Levels	Trends	Levels	Trends	Levels	Trends		
Controls in:								
1. Electricity	0.10*** (0.02)	0.10*** (0.02)			0.39	0.29	0.00	6082
2. Radio	0.10*** (0.03)	0.06* (0.03)	0.10*** (0.03)	0.06* (0.03)	0.26	0.45	0.54	6085
3. Television	0.04** (0.02)	0.04** (0.02)	−0.02 (0.01)	−0.02 (0.01)	0.68	0.87	0.11	6085
4. Refrigerator	0.01** (0.01)	0.01*** (0.01)	0.00 (0.00)	0.00 (0.00)	0.75	0.65	0.00	6084
5. Piped Water	0.05** (0.02)	0.04* (0.03)	0.03 (0.02)	0.03 (0.03)	0.93	0.93	0.05	6086

Notes: This table shows the average treatment effects (ATT) based on [Borusyak et al. \(2024\)](#) for the time frame $T = [0, 6]$. Unit of observation is the household, unit of treatment is the locality; there are 201 localities electrified between 2008 and 2017. Each cell represents a separate estimation model and shows the average treatment effect of electrification on the dependent variable. The columns entitled “Total” calculate the average treatment effects for all households, while the columns entitled “Spillover” calculate the average treatment effects only for households who do not have access to electricity. The dependent variables are all binary, taking the value 1 if the household owns (at least one of) the item. “Piped water” equals one if the household has access to piped water either privately in the household or publicly in the vicinity (i.e. a public standpipe). The community controls are all significant variables from column 5 of [Table 1](#). They are included either in levels or in levels and trends (as specified in the column header). Household controls comprise the sex and age of the household head (plus age squared), as well as the share of adult women in the household who have any primary or secondary education, respectively. Pre-trend p-values refer to the F-test for joint significance of the 6 periods preceding treatment. Standard errors, clustered at the locality level, in parentheses. See Web Data Appendix for data sources.

*** $p \leq 0.01$, ** $p \leq 0.05$, * $p \leq 0.1$.

the treatment group and localities electrified in 2015–2017 represent the comparison group. OLS results are provided as a benchmark. The instrument is again the log distance to the minimum distance spanning tree (MST). Panel A of [Table 5](#) shows the treatment effects for the full sample, while Panel B shows the spillover effects on non-electrified households. We find a large increase in the ownership of mobile phones, regardless of whether the household has an electric connection. This is what one expects if community electrification improves the opportunities for mobile phone charging for everyone. Importantly, we also find a large increase in the share of households that have a bank account. We cautiously interpret this as an improvement in financial inclusion. Mobile phones have the potential to give individuals in remote areas with limited infrastructure access to financial services ([Mothobi and Grzybowski, 2017](#)). The FinScope Burkina Faso Consumer Survey (2016) found that only 40% of adults have access to services from any kind of formal financial institution such as banks and other regulated organizations, 21% rely purely on informal financial services, and 39% are completely financially excluded ([FinMark Trust, 2016](#)). Access in our rural and peri-urban sample is significantly lower still. Part of the problem is the physical inaccessibility of financial service points. The vast majority of adults do not know where the nearest bank, ATM, or microfinance institute is and do not use them. These service points tend to be only accessible by car or motorbike, particularly in rural areas. In contrast, mobile money agents are more easily accessible and tend to be in walking distance. Indeed, 30% of adults used mobile money services in 2016 ([FinMark Trust, 2016](#)). Thus, increased access to mobile phones through community electrification has the potential to improve financial inclusion, particularly for individuals that do not have physical access to traditional financial service points.

6. Discussion and conclusion

In this paper we studied Burkina Faso’s grid extension. The rich and detailed information on planning helped us to shed light on likely sources of endogeneity. The government explicitly targeted more developed localities in terms of their own population, their catchment area and existing private and public infrastructures. Grid supply followed assumed demand. This is probably a typical setting in developing countries. It would let us expect positively biased estimates of the impact when comparing electrified against non-electrified localities.

We argued that the timing of electrification is more exogenous, because grid expansion is bound by engineering considerations. Expanding the grid means that localities closer to the existing network will be connected earlier. Costs are reduced by minimizing total network length. We showed that those engineering variables can predict the timing of electrification well, not only trumping but rendering economic variables insignificant. This motivated us to choose not-yet electrified localities as comparison group and a staggered DiD approach as preferred identification strategy. As a robustness check, we also employ an instrumental variable strategy based explicitly on these engineering considerations.

Our evidence draws a more positive picture of the effectiveness of electrification. In line with much of the literature ([Burlig and Preonas, 2024](#); [Min et al., 2013](#); [Min and Gaba, 2014](#); [Dugoua et al., 2018](#)), we found a significant increase in luminosity after a community got connected to the grid. If, as argued by [Min et al. \(2013\)](#), luminosity reflects mostly street lights, our results suggest significant public investment into street lighting. Of course, luminosity may also reflect private economic activity, household wealth and GDP more broadly ([Henderson et al., 2012](#); [Hodler and Raschky, 2014](#)). In fact, our finding of increases in the household ownership of certain assets underlines this possibility. We found positive effects on public goods that are complementary to electricity. The arrival of the grid enabled the electrification of schools and grid-powered drinking water systems.

However, we also found a number of non-results contradicting claims of the transformative power of electrification. We did not find significant effects on enrollment rates, children’s nutrition and health status, possibly because health inputs did not improve substantially either. Indeed, analyzing various dimensions of public investments, we find it unlikely that electrification was bundled with other development expenditures.

Our findings highlight the importance of considering the extensive margin (the “arrival of the grid”) rather than attempts to increase the electricity uptake of households in already electrified communities (the intensive margin). This is particularly important when positive spillovers exist at the community level. In our context, this was the case. With the arrival of the grid, more households own electric appliances that require electricity such as radios, TVs, refrigerators and mobile phones. Some of these effects also extend to households that do not have their own electricity connection. We find particularly

Table 4
Community effects, Alternative identification strategies.

Treatment group: Community Electrified 2008–2009 Control group: Community Electrified 2010–2014								
Pre-Treatment period	2002–2007 2010–2014			None 2010–2014				
Post-Treatment period	DiD: BJS	DiD: TWFE	Nt/N	OLS	Reduced form	IV: 2SLS	Montiel–Pflueger F	N
1. Luminosity, Extensive margin	0.267*** (0.036)	0.396*** (0.046)	324/3267	0.348*** (0.053)	−0.084*** (0.030)	0.455*** (0.158)	18.96	1376
2. Luminosity, Intensive margin	0.214** (0.106)	0.511*** (0.086)	240/1577	0.299* (0.160)	−0.181** (0.092)	0.959* (0.575)	10.83	665
3. New grid-Powered AEPS/PEA water systems	0.030** (0.013)	0.022 (0.017)	324/3285	0.025* (0.014)	−0.008 (0.006)	0.042 (0.036)	19.07	1380
4. N electrified schools †	0.245*** (0.090)	0.164 (0.128)	216/2733	0.259*** (0.098)	−0.102** (0.045)	0.551** (0.266)	18.89	828
5. N students in public primary schools †	−4.105 (5.468)	−7.481 (5.304)	216/2733	−4.085 (7.072)	−3.368 (3.346)	18.172 (19.143)	18.89	828
6. Annualized population growth (%)	–	–	–	1.004 (1.233)	−0.739 (0.499)	4.010 (2.625)	17.71	272

Notes: Standard errors in parentheses, clustered at the locality level. Column (DiD: BJS) reports Average Treatment Effects on the Treated (ATT) for $T = [0, 6]$ (and $T = [0, 3]$ for those variables marked with a dagger †), using the imputation estimator of [Borusyak et al. \(2024\)](#). Region fixed effects are used instead of locality fixed effects due to sample size constraints. Column (2) reports the canonical TWFE estimator. Column (3) reports the number of treated units (N_t) across all horizons and the total number of observations (N) used in the estimation of DiD BJS. Column (4) reports baseline estimates. Column (5) reports the reduced form estimates using log distance to planned grid poles as instrumental variable when controlling for log distance to development poles and the 33 kV distribution network of 2007 as controls. Column (6) reports 2SLS estimates. Column (7) reports F-statistics for weak instruments according to Kleibergen–Paap and [Montiel Olea and Pflueger \(2013\)](#). The final column reports number of observations in estimations columns (4)–(6). See Web Data Appendix for data sources.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5
Effects of community electrification on household asset ownership, Alternative estimation strategies.

Treatment group: Community Electrified 2008–2009 Control group: Community Electrified 2015–2017							
Pre-treatment period	1993 & 2003		None				Mean (pre)
Post-treatment period	2010 & 2014		2010 & 2014				
Dependent variable	DiD: BJS	DiD: TWFE	OLS	Reduced Form	IV: 2SLS	Montiel–Pflueger F	
1. Electricity	0.17*** (0.04)	0.20*** (0.04)	0.20*** (0.04)	−0.05* (0.03)	0.21** (0.11)	16.38	0.01
2. Radio	0.05 (0.05)	0.08 (0.05)	0.06 (0.06)	−0.07*** (0.02)	0.31** (0.12)	16.40	0.52
3. Television	0.09*** (0.03)	0.09** (0.04)	0.10*** (0.03)	−0.04** (0.02)	0.18* (0.09)	16.38	0.03
4. Refrigerator	0.03** (0.01)	0.03** (0.01)	0.01 (0.01)	−0.01 (0.01)	0.02 (0.03)	16.38	0.00
5. Piped Water	0.05 (0.04)	0.05 (0.06)	−0.03 (0.04)	0.03 (0.02)	−0.13 (0.09)	16.40	0.05
Number of households (N)	3232	3232	1866	1866	1866		
Number of localities (T/C)	101 (26/75)	101 (26/75)	69 (17/52)	69 (17/52)	69 (17/52)		

Notes: Unit of analysis is the household, unit of treatment is the locality. The first two columns show difference in difference estimates based on a reduced sample of observations, where we consider only localities electrified pre 2010 or post 2014. This sample creates a 2×2 matrix that allows for a classic two-way fixed effects (TWFE) DiD model. We show results using BJS estimation ([Borusyak et al., 2024](#)) on this reduced sample, alongside the TWFE results. The next three columns show estimates based on cross-sectional data from 2010 and 2014: ordinary least squares, and IV estimates including reduced form and two-stage least squares (2SLS). The last two columns report F-statistics testing for weak instruments, according to [Montiel Olea and Pflueger \(2013\)](#), and the mean of the dependent variable in the pre-treatment period. All estimations control for community-level characteristics (pre-treatment) and region fixed effects. Household-level controls are not included, as these would not be valid for the IV specification (and models should be directly comparable). Standard errors, clustered at the locality level, in parentheses. See Web Data Appendix for data sources.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

strong increases in the ownership of mobile phones and radios, both of which are typically battery operated and do not require a constant electric connection. We did not find spillovers in the ownership of TVs and refrigerators, which require a constant source of power. This is suggestive of the possibility that electricity is sold on in the community, with connected households giving temporary access to their neighbors. Indeed, in the 2016 FinScope Survey over a third of adults who own a mobile phone reported charging it at a shop or kiosk close by and 8% reported charging it at a neighbor's house. [Peters and Sievert \(2016\)](#) find that across rural settings of sub-Saharan Africa, the most common

income-generating use of household electricity consists of providing services to the local community, such as mobile phone charging or “cinemas”. “Similarly, [Lenz et al. \(2017\)](#) observe that in Rwanda, grid-connected households often rent out mobile phones to display films or football matches for a fee, while those without connections save on batteries, kerosene, and mobile phone charging costs by benefiting from informal sharing in electrified areas.

We also demonstrated that electrification increases the proportion of households with bank accounts, including for unconnected households. Financial exclusion remains a large problem in Burkina Faso.

Table 6

Household access to mobile phones and banking - Cross-Sectional estimation of the effect of community electrification.

Cross-Sectional analysis using data from 2010, 2014 Treatment group: Community Electrified 2008–2009 Control group: Community Electrified 2015–2017						
Panel A: Total effects						
Dependent variable	OLS	Reduced form	IV: 2SLS	Montiel–Pflueger F	Mean (Control)	N
1. Mobile phone	0.03 (0.04)	−0.04*** (0.01)	0.19** (0.09)	16.04	0.67	1862
2. Bank account	0.06* (0.03)	−0.04* (0.02)	0.16** (0.09)	15.93	0.08	1845
Panel B: Spillover effects						
Dependent variable	OLS	Reduced form	IV: 2SLS	Montiel–Pflueger F	Mean (Control)	N
1. Mobile phone	0.00 (0.04)	−0.04*** (0.01)	0.18* (0.10)	12.01	0.67	1744
2. Bank account	0.02 (0.04)	−0.04** (0.02)	0.18** (0.09)	11.85	0.08	1730

Notes: Unit of analysis is the household, unit of treatment is the locality. OLS, reduced form, and 2SLS estimations are based on 69 localities, 17 of which are in the treatment group. The dependent variables are all binary, taking the value 1 if the household owns at least one of the item. All estimations control for household and community-Level Controls and region fixed effects. The last three columns report F-statistics testing for weak instruments (Montiel Olea and Pflueger, 2013), the mean of the dependent variable in the control group, and the number of observations (households). Standard errors, clustered at the locality level, in parentheses. See Web Data Appendix for data sources.

*** p<0.01, ** p<0.05, * p<0.1.

As mobile money services have been expanding rapidly (N'dri and Kakinaka, 2020), ICTs become more important (Karakara and Osabuo-hien, 2019). The fact that mobile phone ownership and bank accounts simultaneously expand in electrified localities can be seen in this light.

We found suggestive evidence that electrified localities may have outgrown not-yet electrified localities in terms of population. Most likely, this result stems from migration from never-electrified localities (outside our sample). People probably will not move to essentially similar localities in order to benefit from a slightly earlier date of electrification. We would rule out natural growth. First, we did not find large measurable effects on child mortality. Second, we would expect, if at all, a negative effect of electrification on fertility (Akpanjar and Kitchens, 2017; Grimm et al., 2015; Grogan, 2016).

Overall, this paper has demonstrated that there is an important community dimension to the benefits of electrification, comprising both the provision of public goods, as well as spillover effects from electrified households to the community. The community dimension is well recognized for other infrastructure investments, such as roads and water provision, but the literature studying the welfare effects of electrification has so far focused mainly on the private benefits of household electric connections at the intensive margin. An accurate assessment of the welfare effects of grid extension needs to take into account public goods at the community level and spillover effects.

CRedit authorship contribution statement

Maika Schmidt: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Alexander Moradi:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Appendix A. Supplementary materials

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jdeveco.2025.103556>.

Data availability

Replication package: Harvard Dataverse (DOI: <https://doi.org/10.7910/DVN/WOORX>).

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Does Africa Need a Rotten Kin Theorem?

Experimental Evidence from Village Economies

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Abstract

This paper measures the economic impacts of social pressures to share income with kin and neighbors in rural Kenyan villages. We conduct a lab experiment in which we randomly vary the observability of investment returns to test whether subjects reduce their income in order to keep it hidden. We find that women adopt an investment strategy that conceals the size of their initial endowment in the experiment, though that strategy reduces their expected earnings. This effect is largest among women with relatives attending the experiment. Parameter estimates suggest that women anticipate that observable income will be “taxed” at a rate above four percent; this effective tax rate nearly doubles when kin can observe income directly. Though this paper provides experimental evidence from a single African country — Kenya — observational studies suggest that similar kin pressures may be prevalent in many rural areas throughout Sub-Saharan Africa.

JEL codes: C91, C93, D81, O12

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1 Introduction

Risk is a pervasive aspect of the lives of individuals in many developing economies, and informal risk-pooling arrangements which help households cope with shocks can have substantial welfare impacts when credit and insurance markets are incomplete. A substantial body of empirical evidence documents the existence of mutual insurance arrangements throughout the world, demonstrating that informal mechanisms typically fail to completely insure households against idiosyncratic shocks (cf. Townsend 1994, Coate and Ravallion 1993, Fafchamps and Lund 2003). Much of the literature focuses on mutual insurance arrangements which are efficient given constraints, characterizing the conditions under which self-interested households will enter risk-pooling schemes voluntarily *ex ante* and the participation constraints which keep households from defecting *ex post*.¹ However, the expectation of future transfers is only one of many reasons households offer assistance to those worse off: altruism, guilt, and social pressure to share income may also play a role (Scott 1976, Foster and Rosenzweig 2001, Alger and Weibull 2010). In fact, several recent studies suggest that individuals living in poor, rural communities often feel obligated to make transfers to relatives and neighbors, and that successful families who do not make sufficient transfers to others can face harsh social sanctions (Platteau 2000, Hoff and Sen 2006, Di Falco and Bulte 2011, Comola and Fafchamps 2011, Dupas and Robinson forthcoming-b). For example, Barr and Stein (2008) argue that Zimbabwean villagers punish households who are becoming better off than their neighbors by refusing to attend the funerals of members of those families. When social pressures to assist kin, and the sanctions against those who violate sharing norms, are strong enough, they can reduce incentives to make profitable investments and drive savings into lower-return technologies which are less observable to family members. Baland, Guirking, and Mali (2011) provide evidence of this type of behavior in Cameroon, where members of credit cooperatives take out loans to signal that they are liquidity constrained — even when they also hold substantial savings — in order to avoid sharing accumulated wealth with relatives. Anecdotal evidence suggests that pressures to share income with kin are common in many parts of the world. However, Collier and Garg (1999)

¹See Ligon, Thomas, and Worrall (2002), Albarran and Attanasio (2003), and Kinnan (2011) for examples. Foster and Rosenzweig (2001), which explores the impact of altruism on the set of self-enforcing insurance arrangements, is an important exception.

argue that such pressures are most intense in Africa, and are common to many ethnic groups across the continent; they write that “African kin groups are distinctive both by their ubiquity and by the strength of their claims upon members.”

In this paper, we report the results of an experiment designed to measure social pressure to share income with relatives and neighbors in rural villages in Sub-Saharan Africa. We use a controlled laboratory environment to explore behaviors which are difficult or impossible to document using survey data: the willingness to forgo profitable investment opportunities to keep income secret. We conduct economic experiments in 26 rural, agricultural communities in western Kenya. Within the experiment, subjects receive an endowment which they divided between a risk-free savings account and a risky but profitable investment. The size of the endowment varies across subjects, and the distribution of endowment sizes is common knowledge. While the amount saved is always private information, we randomly vary whether the amount invested in the risky security can be observed by other subjects, creating an incentive for those receiving the large endowment to invest no more than the amount of the small endowment, thereby keeping their endowment size hidden. We also offer a subset of subjects the option of paying to keep their investment returns secret, allowing us to directly measure the willingness-to-pay to hide income.

In light of evidence that women and men have different risk preferences (cf. Croson and Gneezy 2009) and that women in poor communities may have more trouble accumulating savings (cf. Dupas and Robinson forthcoming-a), we stratify our experiment by gender, and report results for men and women separately.² We find convincing evidence that women are willing to reduce their expected income to avoid making investment returns observable. Women receiving the large endowment, who may wish to hide this fact from their relatives and neighbors, are 25.4 percent (9.6 percentage points) more likely to invest an amount no larger than the small endowment when returns are observable; this is equivalent to a 5.4 percentage point decline in investment level. We find no similar tendency to hide income among men. The effect we observe among women appears to be driven primarily by the behavior of those with relatives attending the experiment, who would

²Dupas and Robinson (forthcoming-a) find evidence that female daily wage earners in western Kenya are more savings constrained than men in similar occupations. In a similar vein, De Mel, McKenzie, and Woodruff (2008), De Mel, McKenzie, and Woodruff (2009), and Fafchamps, McKenzie, Quinn, and Woodruff (2011) find lower returns to capital for microenterprises operated by women than for those operated by men.

2 Experimental Design and Procedures

2.1 Structure of the Experiment

The experiment was designed to introduce exogenous variation in the observability of investment returns. Within the experiment, each participant was given an initial endowment, either 80 or 180 Kenyan shillings.⁴ Each subject divided her endowment between a zero-risk, zero-interest savings account and an investment which was risky but profitable in expectation. The subject received five times the amount that she chose to invest in the risky prospect with probability one half, and lost the amount invested otherwise. A coin was flipped to determine whether each risky investment was successful. Thus, the main decision subjects faced was how much of their endowment to invest in the risky security and how much to allocate to the secure, zero-profit alternative.

Within the experiment, players were randomly assigned to one of six treatments. First, players were allocated either the smaller endowment of 80 shillings or the larger endowment of 180 shillings. Endowment sizes were always private information — experimenters never identified which subjects received the large endowment. However, the distribution of endowments was common knowledge, so all subjects were aware that half the participants received an extra 100 shillings.

Every player was also assigned to either the **private** treatment or one of two public information treatments, the **public** treatment or the **price** treatment. Participants assigned to the **private** treatment were able to keep their investment income secret: the decisions they made in the experiment were never disclosed to other participants. In contrast, those assigned to the **public** treatment were required to make an announcement revealing how much they had invested in the risky security, and whether their investment was successful, to all of the other participants at the end of the experiment.⁵ The amounts that subjects invested in the zero-interest savings technology were never revealed, so those who received the larger amount could choose whether to invest

⁴Experimental sessions were conducted between August 3 and October 1, 2009. Over that period, the value of the US dollar relative to the Kenyan shilling fluctuated between 74.5 and 77.25 shillings to the dollar. We report dollar amounts using the exchange rate 75.9 shillings to the dollar, which is the average over that period. The two endowments were equivalent to 1.05 and 2.35 U.S. dollars, respectively. Among our subjects, the median monthly wage for individuals in full-time, unskilled employment is 2000 shillings (26.35 U.S. dollars), and the median monthly wage for subjects in full-time work in the skilled sector is 4750 shillings (62.58 U.S. dollars).

⁵Subjects were informed up front that they were allowed to delegate the task of making the public announcement to a member of the research team if they wished to avoid the public speaking aspect of the announcement process.

80 shillings or less so as to obscure their endowment size. Finally, those assigned to the **price** treatment were obliged to make the public announcement revealing their investment returns unless they preferred to pay a price, p , to avoid making the announcement. Prices ranged from 10 to 60 shillings, and were randomly assigned to subjects in the price treatment. Subjects were informed what price they faced *before* making their investment decisions, but decided whether to pay the price *after* investment returns were realized. Hence, subjects in the price treatment were not always able to afford to buy out: those who invested and lost a large fraction of their endowment did not always have enough experimental income left over to pay the exit price, p .⁶

Random assignment to treatment generated exogenous variation in the observability of investment returns and created costly opportunities to hide income. Assignment to the public information treatments meant that outcomes were verifiable, and might therefore facilitate risk-pooling and, consequently, risk-taking. On the other hand, if subjects face social pressure to share income with neighbors and kin, they might be willing to pay for obscurity when returns are visible.⁷ In particular, the experiment creates two mechanisms through which subjects could incur a cost to hide income from others. First, those receiving the larger endowment could keep their endowment size secret by investing no more than 80 shillings. Second, subjects in the price treatment could pay the randomly assigned price, p , to conceal their income entirely.

2.2 Experimental Procedures

Experiments were conducted in 26 rural, predominantly agricultural communities in western Kenya.⁸ One day prior to each experimental session, the survey team conducted a door-to-door recruitment campaign, visiting as many households within the village as possible.⁹ All households within each village were invited to send members to participate in the experimental economic game session the

⁶Subjects were never allowed to use money from outside the experiment to pay to avoid the public announcement.

⁷Our design is related to those of Barr and Genicot (2008), Attanasio, Barr, Cardenas, Genicot, and Meghir (2012), and Ligon and Schechter (forthcoming), who also use experiments which capture the desire to avoid social sanctions outside the lab.

⁸Communities were selected to be at least five kilometers apart from one another, to prevent overlap in subject populations, and to avoid areas where IPA–Kenya had ongoing projects.

⁹When the recruitment team was unable to contact and survey a household prior to the experiment, village elders were asked to invite them to attend the subsequent experimental session.

following day. 80.4 percent of households contacted prior to the sessions chose to participate.¹⁰ Experimental sessions were conducted in empty classrooms at local primary schools. Sessions included an average of 83 subjects; no session included fewer than 65 or more than 100 subjects. Each session lasted approximately three hours.

Within each session, participants were stratified by gender and education level (an indicator for having done any post-primary education). There were six experimental treatments, corresponding to the three information conditions (private, public, price) interacted with the two endowment sizes. Within each stratum, players were randomly assigned to each of the six treatments with equal probability.¹¹ Players assigned to the price treatments were subsequently assigned a random price from the set of multiples of ten between 10 and 60.¹²

Experimental sessions were structured as follows. After a brief introduction, enumerators read the instructions and answered participant questions, illustrating the decisions that a subject might face with a series of wall posters.¹³ Subjects were then called outside one at a time, by ID number, to make their investment decisions. Since some participants had limited literacy skills, decisions were recorded by members of the research team. To ensure that earnings not announced publicly remained private information, each enumerator sat at a desk in an otherwise empty section of the schoolyard. Enumerators began by asking a series of questions designed to make sure that subjects understood the experiment. Subjects were then informed whether they had received the large or small endowment and whether they were assigned to the private, public, or price treatment. Those who were assigned to the price treatment were also told what price they would need to pay if they wished to avoid the public announcement. Subjects then made their investment decisions: each subject was handed a number of 10 shilling coins equivalent to her endowment; the participant

¹⁰In Table of Appendix III (online), we examine the correlates of choosing to send a household member to the experimental session. Participants were broadly representative of the overall population, though as expected larger households were more likely to send a member to the experiment. Women from poorer households and those with relatively higher math skills were more likely to attend. Men were more likely to attend if they attended church the previous week. Subjects in more isolated communities — as measured by distance from a paved road — were more likely to participate, though this cannot bias our results since assignment to treatment occurs within villages.

¹¹The first experimental session did not include the two price treatments, so players were assigned to each of the other four treatments with equal probability.

¹²Note that each session included subjects assigned to all three information conditions, so it was impossible to infer which individuals chose to pay to avoid the public announcement, since these subjects could not be distinguished from those randomly assigned to the private treatments.

¹³Detailed instructions are included in Appendix II (online).

divided these coins between a “savings” cup and a “business” cup.¹⁴ After recording a subject’s investment decision, the enumerator would give the subject a one shilling coin to flip.¹⁵ The outcome of the coin flip determined whether the money placed in the business cup was multiplied by five or removed from the subject’s final payout. Subjects assigned to the price treatment were then asked whether they wanted to pay the fee to avoid announcing their investment results. If they had enough money left to pay the fee, and they chose to do so, it was deducted from their payoff. After all decisions had been recorded, public announcements were made. At the end of the session, subjects were called outside one at a time to receive their payouts. Each subject received her payout in private, and was allowed to leave immediately after receiving her money. Figure 1 summarizes the progression of activities within the experiment.

2.3 Experimental Subjects

Sessions were conducted in Kenya’s Western Province, in three adjoining districts: Bunyala, Samia, and Butula.¹⁶ All three districts are predominantly smallholder farming communities, though Samia and Bunyala also have ports on Lake Victoria. Summary statistics on experimental subjects are presented in Table 1. 61 percent of subjects were female. Respondents ranged in age from eighteen to 88. 9.4 percent of subjects had no formal schooling, while 12.2 percent had finished secondary school. The median participant was a 34 year old married woman with seven years of education, living in a six-person household with her husband and her four children. The median participant’s household owns one bicycle, one cell phone, four chickens, and two mosquito nets, but does not own a television, any cattle or goats, or a motor vehicle. 23.0 percent of respondents live in households with at least one employed household member; most (64.6 percent) employed subjects do agricultural work or other unskilled labor. The median monthly wage among participants with full-time employment was 2,050 Kenyan shillings, or 1.35 USD per day (assuming twenty work

¹⁴The swahili words for “savings” and “business” are, respectively, “akiba” and “biashara.” The savings cup was marked with a large letter “A” and the business cup was marked with a large letter “B.” The two cups made of plastic and were identical in size and color.

¹⁵To limit the possibility of influencing the outcome of the coin flip, each subject placed the coin into a sealed, opaque container which she shook vigorously before opening it to reveal the outcome of the coin toss.

¹⁶Kenya’s recent redistricting carved these three former administrative divisions of Busia District off as new districts of their own. One of the districts, Butula, was declared a new district during the course of this project.

employment. Thus, stakes were large, but not life-altering. There is substantial variation in payoffs across across treatments: subjects in the public, small endowment treatment received the lowest payoffs, averaging 139 Kenyan shillings (1.83 US dollars); those in the private, large endowment treatment earned the most, on average 355 Kenyan shillings (4.68 US dollars).

4.1 Individual Investment Decisions

On average, subjects chose to invest just over half their endowment in the business cup (Table 2). The fraction of the endowment invested is similar in the large and small endowment treatments. Among those receiving the larger endowment, the amount invested is slightly lower in the two public information treatments (the public and price treatments) than in the private treatment. When allotted 180 shillings, subjects could avoid revealing that their endowment exceeded that of others by investing 80 shillings or less; the probability of doing so is higher in the public information treatments than in the private treatment.

We present histograms of investment decisions in Figure 4. Dark bars represent the distribution of investment decisions in the private treatments; lighter bars represent investment decisions in the two public information treatments. As predicted by the model, decisions are similar in the public and private small endowment treatments. However, women receiving the large endowment are more likely to invest exactly 80 shillings in the public information treatments than in the private, large endowment treatment. A Kolmogorov-Smirnov test rejects the hypothesis that the two distributions are equal (p-value 0.068). There is no difference in behavior across treatments among men receiving the large endowment.

We now examine the impact of observability on investment decisions in a regression framework, estimating the effects of assignment to one of the public information treatments on the amount of money invested in the business cup. We estimate the OLS regression:

$$Investment_{itv} = \alpha + \beta_1 Public_i + \beta_2 Large_i + \beta_3 Public \times Large_i + \eta_v + X_i' \zeta + \varepsilon_{itv} \quad (14)$$

where $Investment_{itv}$ is the amount invested in the business cup by subject i assigned to treatment

t in village v , $Public_i$ is an indicator for assignment to either the public or the price treatment, $Large_i$ is an indicator for receiving the large endowment, $Public \times Large_i$ is an interaction between $Public_i$ and $Large_i$, η_v is a village fixed effect, and X'_i is a vector of individual controls including dummies for age and education categories, marital status, household size, and the log value of household durable assets.²⁷ Standard errors are clustered at the village level in all specifications.

Results, disaggregated by gender, are reported in Table 3. Odd-numbered columns report regression results without controls; even-numbered columns include them. The $Public \times Large_i$ interaction is negative and significant among women (Columns 1 and 2), but positive and insignificant among men (Columns 3 and 4). The coefficient estimates indicate that women receiving the large budget invest approximately 6.3 shillings, or 6.5 percent, less when returns are observable than when they are hidden. Adding village fixed effects and additional controls has almost no impact on estimated coefficients and significance levels.

Next, we test whether the observed changes in investment levels are consistent with women attempting to obscure the size of their endowments. We test the hypothesis, predicted by the model, that subjects are more likely to invest exactly 80 shillings when investment returns are observable by estimating probit regressions of the form

$$\Pr [Investment = 80] = \Phi (\alpha + \beta Public_i + X'_i \zeta) \quad (15)$$

among the sample of subjects randomly assigned to the large endowment treatments, as well as a linear probability model (LPM) where the dependent variable is an indicator for investing exactly 80 shillings (Table 4, Panel A). In all specifications, women are significantly more likely to invest exactly 80 shillings in the public information treatments. For example, the LPM estimates indicate that women are 6.2 to 7.0 percentage points more likely to invest 80 shillings when investment returns are observable. The coefficient on $Public$ in the sample of men is not significant in any specification, in line with our previous results.

²⁷As discussed in Section 3, decisions in the price treatments will be identical to decisions in the public treatments when p is sufficiently high, so we pool data from both public information treatments to maximize power. Coefficient estimates are similar when data from the price treatments is omitted.

Moving beyond the precise predictions of the model presented in Section 3, we also estimate similar specifications using indicators for investing either 70 or 80 shillings (Table 4, Panel B) and for investing any amount less than or equal to 80 shillings (Table 4, Panel C). Since only 2.3 percent of women and 6.0 percent of men in the private, small endowment treatment actually invest 80 shillings, subjects might feel that investing 70 shillings better obscures the size of their endowment.²⁸ In more general terms, any investment level less than or equal to 80 shillings obscures the size of the subject's endowment, and may be seen as consistent with a desire to hide a portion of the large endowment from others. In all specifications, the *Public* variable is positive and significant in the sample of women, but insignificant in the sample of men. Women receiving the large endowment are between 7.0 and 7.6 percentage points more likely to invest 70 or 80 shillings and between 9.6 and 10.5 percentage points more likely to invest 80 shillings or less than those in the private, large endowment treatment.

4.1.1 Treatment Effect Heterogeneity

Our hypothesis is that women face social pressure to share income, and this creates an incentive to hide investment returns when possible, even if it is costly to do so. Under this hypothesis, the extent of income hiding should be associated with factors predicting the level of social pressure an individual is likely to face after the experiment. We focus on close kin outside the household as the group most likely to pressure individuals into sharing income.²⁹ We test this by interacting the *Public* variable with indicators for whether or not a subject had any close kin attending the experiment, thereby disaggregating the impact of the public information treatments.³⁰ We estimate

²⁸Investing less than, but not exactly 80 shillings may reflect level-k reasoning, along the lines first discussed by Nagel (1995), Stahl and Wilson (1994), and Stahl and Wilson (1995).

²⁹Hoff and Sen (2006) highlight the role played by kin networks in extracting surplus from successful relatives, while Baland, Guiringer, and Mali (2011) provide evidence that individuals seek to hide income from family members.

³⁰We define close kin as parents, grandparents, siblings, grown children, and aunts and uncles. Note that we were unable to stratify treatment assignment by kin presence because the variable was constructed after experimental sessions took place using a name-matching algorithm.

the OLS regression equation

$$Y_{itv} = \alpha + \beta_1 KinPresent_i + \beta_2 Public \times KinPresent_i + \beta_3 Public \times NoKinPresent_i + \eta_v + X_i' \zeta + \varepsilon_{itv} \quad (16)$$

where Y_{itv} is one of three outcomes of interest: investment level, an indicator for investing exactly 80 shillings, or an indicator for investing no more than 80 shillings.³¹ We restrict the sample to subjects assigned to the large endowment treatments, who could obscure the size of their endowment by investing 80 shillings or less. We include village fixed effects (η_v) and controls for age and education categories, HH size, marital status, and household durable asset holdings in all specifications.

Results are reported in (Table 5, Panel A). Women with close kin attending the experiment invest less in the public information treatments and are more likely to invest exactly 80 shillings and no more than 80 shillings. The coefficient estimates are extremely large. Women with relatives present invest 24.5 shillings less (p-value 0.0175) when investment returns are observable than when they are hidden, which is equivalent to more than a twenty percent reduction in investment relative to the private information treatment. These women are also 14.9 percentage points more likely to invest exactly 80 shillings (p-value 0.0491) and 41.8 percentage points more likely to invest 80 shillings or less (p-value 0.0012). In contrast, point estimates are smaller and not statistically significant for women with no kin attending the experiment. As expected, we also find no statistically significant results for men.

An alternative hypothesis is that kin are significant because they pass information about wives' incomes to their husbands.³² This is of particular concern because 76.5 percent of women in our sample are married, and the two main ethnic groups in the area — the Luhya and the Luo — are both patrilocal (Brabin 1984, Luke and Munshi 2006), so married women typically live near their in-laws rather than their blood relatives. We present two pieces of evidence which suggest that kin are not simply conveying information to husbands.

³¹We estimate linear probability models for the two binary outcomes. Probit results are similar in magnitude and significance, and are omitted to save space.

³²Robinson (2008) and Ashraf (2009) document limited commitment and observability effects *within* the household, while Anderson and Baland (2002) argue that Kenyan women use ROSCAs to hide savings from their husbands.

on exogenous factors: the randomly assigned price and indicators for receiving the large endowment and having the coin land with heads facing up. Women with more observable income are clearly more likely to pay to avoid the public announcement: the coefficient on *Heads* is consistently positive and significant, suggesting that subjects with successful investments are approximately 23 percentage points more likely to pay to avoid announcing their investment return. Men, in contrast, are not significantly more likely to pay to avoid the announcement when their investments are successful. As suggested by the figures, both men and women are less inclined to buy out at higher exit prices: the coefficient on the price variable is negative and significant in all specifications.

Analysis of factors associated with willingness-to-pay is complicated by the fact that, while prices are randomly assigned, gross payouts (savings plus investment returns) are not. As the model suggests, the relationship between pressure to share income, individual risk preferences, and the unconditional probability of paying to avoid the public announcement is complicated; and we observe whether a subject pays the exit price after a particular realization of the coin flip, but not whether she planned to never buy out, buy out if her investment was successful, or always buy out. In light of this, we defer consideration of the explicit willingness to pay to Section 5.

Table 8 reports the results of regressions of amount invested within the experiment on the price of exit, restricting the sample to those randomly assigned to the price treatments. The model predicts that for subjects planning to pay to avoid the public announcement whether or not their investment was successful, the amount invested will be a decreasing function of the price avoiding the public announcement. Consistent with this prediction, we find that random assignment to a higher price of avoiding the public announcement is associated with significantly lower levels of investment. In contrast to our previous results, this appears to be true for both women (Columns 1 and 2) and men (Columns 3 and 4).

5 Estimating the Social Pressure Parameter

In this section, we estimate the magnitude of the social pressure parameter, τ , while controlling for unobserved heterogeneity in individual risk aversion. As discussed above, the treatment effects

reported in Section 4 demonstrate that among women τ is not equal to zero: making investment returns observable impacts individual choices. However, the magnitude of the reduced form impact does not identify τ because the treatment effect depends on an individual's level of risk aversion. Fortunately, random assignment should insure that the distribution of individual risk parameters in each of the six treatments is representative of the population distribution. We can therefore estimate the parameters of that distribution using data from the private treatments, and estimate τ while controlling for heterogeneity in risk aversion in a structural framework.

We model decisions in a discrete choice setting which mirrors our implementation of the portfolio choice problem in the experiment: each subject decides how many ten shilling coins to allocate to the business cup, and allocates the remainder to the savings cup. Subjects in the small endowment treatments have nine options $(0, 10, 20, \dots, 80)$, while those assigned to the large endowment treatments have nineteen $(0, 10, 20, \dots, 180)$.

In what follows, we index experimental treatments with t , individual subjects with i , and investment options with j and, when necessary, k . We assume that subject i 's expected utility of investing b_j is given by:

$$EU_{ij} = EV_{ij} + \varepsilon_{ij} \quad (17)$$

where EV_{ij} denotes the explicitly-modeled expected utility of investing b_j (what Train (2003) terms “representative utility”) and ε_{ij} is an i.i.d. type 1 extreme value distributed preference shock.³⁶ Behavior in the private treatments suggests that individuals have heterogeneous risk preferences. As in Section 3, we assume these can be represented by a CRRA utility function, $v_i(\cdot)$, parameterized by ρ_i , and that individual ρ_i parameters are normally distributed with mean μ_ρ and variance σ_ρ^2 . The probability that subject i chooses to invest b_j can then be written in the form of a mixed logit model:

$$P_{ij} = \int \left(\frac{e^{EV_{ij}/\sigma_\varepsilon}}{\sum_{k=1, \dots, J_t} e^{EV_{ik}/\sigma_\varepsilon}} \right) f(\rho) d\rho. \quad (18)$$

where σ_ε^2 is proportional to the variance of $\varepsilon_{ij} - \varepsilon_{ik}$.³⁷ Let y_{ij} be an indicator function equal to one

³⁶Loomes (2005) refers to such error terms as “Fechner errors.” See Hey and Orme (1994) and Von Gaudecker, van Soest, and Wengström (2011) for examples of their use in modeling stochastic choices in individual decision-making experiments.

³⁷When $V_{ij} = X' \beta$, β and σ_ε are not separately identified; σ_ε is identified in our framework because EV_{ij} is a

by allowing τ to depend on whether a subject's close relatives were present at the experiment. In Table 11, we report parameter estimates when we replace the single τ parameter with two separate parameters, $\tau_{no\ kin\ present}$ and $\tau_{kin\ present}$. Among women, $\tau_{no\ kin\ present}$ is slightly smaller than in the full sample, at 0.043, but remains statistically significant. The estimated value of $\tau_{kin\ present}$ is 0.080 — substantially higher, and also statistically significant. Among men, as in the full sample, $\tau_{no\ kin\ present}$ remains about 2.7 percent, while the estimated $\tau_{kin\ present}$ is slightly below zero and imprecisely estimated. Thus, consistent with our reduced form results, we find evidence that women with kin present at the experiment are more exposed to social pressure to share income.

5.4 Simulations

5.4.1 Alternative laboratory experiments

The reduced form results depend on the specific parameters of the investment scenario that we presented to the participants, while the structural parameter estimates shown in Tables 10 and 11 do not. To illustrate this, we briefly show how the reduced form results would have differed with the same population, but using slightly different experimental parameters.

To simulate the experiment for women, we use the parameters in Column 2 of Table 10: $\mu_\rho = 0.7488$, $\sigma_\rho = 0.1992$, $\sigma_\varepsilon = 0.0125$, $\tau = 0.045$, and $\gamma = 0.0588$. With these values, we can simulate behavior under a variety of settings. For a simple demonstration, we consider the same laboratory experiment but with different investment returns. In the actual lab setting, as described in Section 2, the investment amount was multiplied by five when successful, and zero when unsuccessful. We simulate the experiment for a range of values of the successful investment return on the interval $[2,12]$, and show the predictions for one of the reduced form regressions — Table 4, Panel C, Columns 3 and 4. For subjects receiving the large endowment treatment, we compare behavior in the private treatment to choices in the two public information treatments, and measure the change in the probability of investing no more than the small endowment (80 shillings).

For the value 5.0, highlighted in Figure 9, the predicted change in the fraction of participants investing 80 shillings or less is 8 percentage points. This is close to the point estimates of 9.6 and 10.5 percentage points shown in Columns 3 and 4 of Table 4, Panel C. Figure 9 shows, however, that

6 Conclusions

We report the results of a novel economic experiment designed to measure social pressure to share income in Kenyan villages. Our approach is stratified to ensure balance, randomized within villages, and is conducted on a large sample. The design permits both reduced-form estimates to find results in line with comparative static predictions, and structural estimates to identify parameters of interest in the presence of heterogeneity.

Women who know that the outcome of their investments will be made public make investments that are less profitable in expectation. Results are strongest for those who have relatives present at the experiment. When we offer some participants the opportunity to pay a fee to avoid making an announcement, they do so at substantial cost: 30 percent of those able to pay to avoid the public announcement choose to do so; these subjects sacrifice 15 percent of their gross payout, on average. Structural estimates of the average “kin tax” are significantly different from zero for women, estimated at roughly 4.3 percent for those whose relatives did not attend the experiment, but at 8.0 percent for those with relatives present. Our model of stochastic choices in the experiment fits the data well, explaining both investment and exit decisions. We see no evidence that this behavior can be explained by household bargaining with a spouse, aversion to public announcements, or aversion to risk-taking in general.

We hypothesize that the behavior observed in this experiment is a sign that village sharing norms distort investment incentives towards less visible, but potentially less profitable, investments, and may consequently slow economic growth. The negative correlations we observe between the extent of income hiding at the village level and the level of prosperity in the village, measured several different ways, are in agreement with this interpretation. However, such results should be interpreted with caution, since the direction of causality is unclear. Moreover, the efficiency impacts of social pressure to share income will clearly depend on the range of income-hiding technologies available, and correlation between observability and profitability. There is still much to be learned about the range of observable and unobservable investment opportunities available to poor households, the set of technologies for hiding or protecting income from pressure to share,